

# THE LEVELING LINE

A3 BOOKLET

## A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.  
Depart, carry the height station by station, return — that gap is f.  
Within tolerance, every station on the line is accepted;  
over tolerance the whole run is void; no station may be patched alone.

That hundred-year-old rule answers this belt's hardest question:  
what a city's trust in its AI rests on: not performance, but return.



### Ground within a ten-minute walk of a benchmark

Dashed: 232 buildings today. Filled: the ground that only comes within ten minutes once the eleven stitching links and three spurs are built — 261 in all, a gain of 29.  
Catchment 203.3 ha to 220.6 ha; 800 m of network distance, not a radius.

READINGS, ALL RECOMPUTED FROM THE SHIPPED FILES

**11,412,825 m<sup>2</sup>**

Overall design area

**13.8 km/km<sup>2</sup>**

Network density, measured

**9,443 m**

Spine length

**17**

Oversized blocks, not parks

**8**

Tiered benchmarks

**19.7%**

Green corridor, share of design area

### What this proposal does not give here matters as much as what it does

Plot ratio, height, coverage, setbacks, road redlines — inside the statutory approval process; none is drafted here before the official conditions are published.

The tolerance values per scenario — set by the tolerance assembly at BM-1; this proposal gives the grading basis and drafts no value.

Any engineering feasibility conclusion — stitching points register a need for connection; transport and structural review is outside this proposal.

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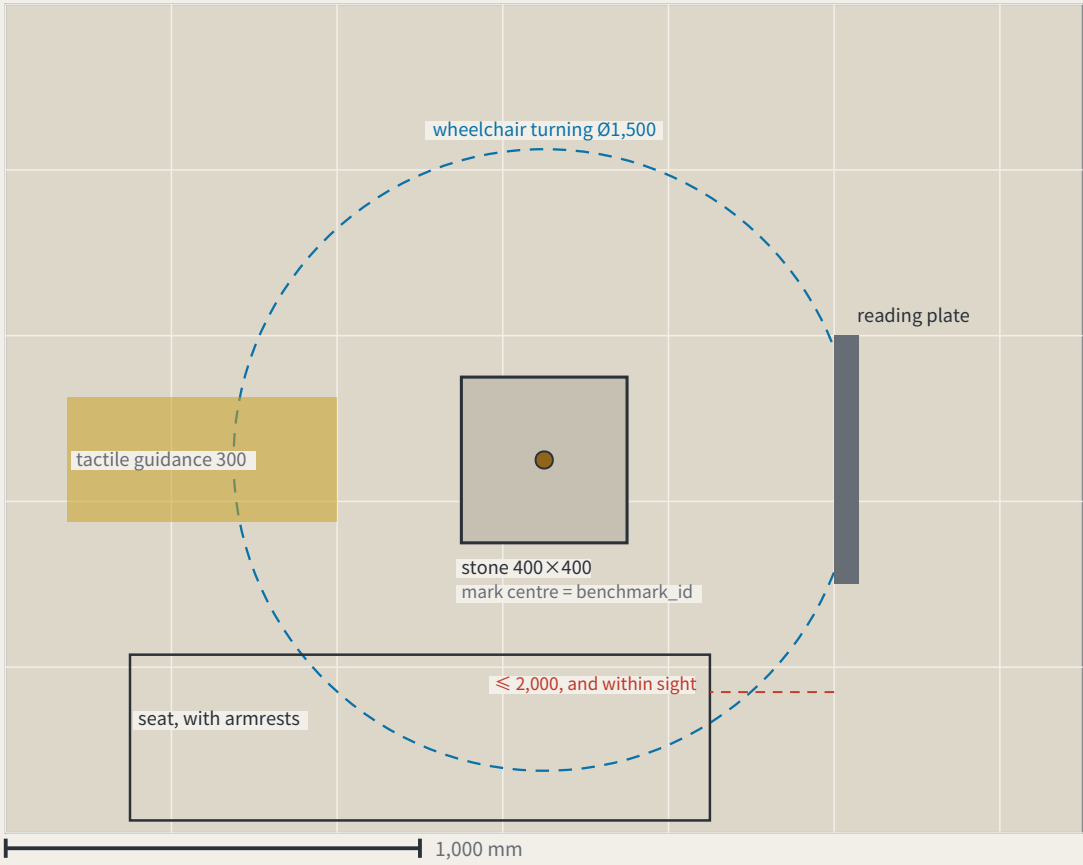
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The benchmark and its reading plate, as a construction detail

FIG.16

a type drawing; it carries no orientation

I. Plan

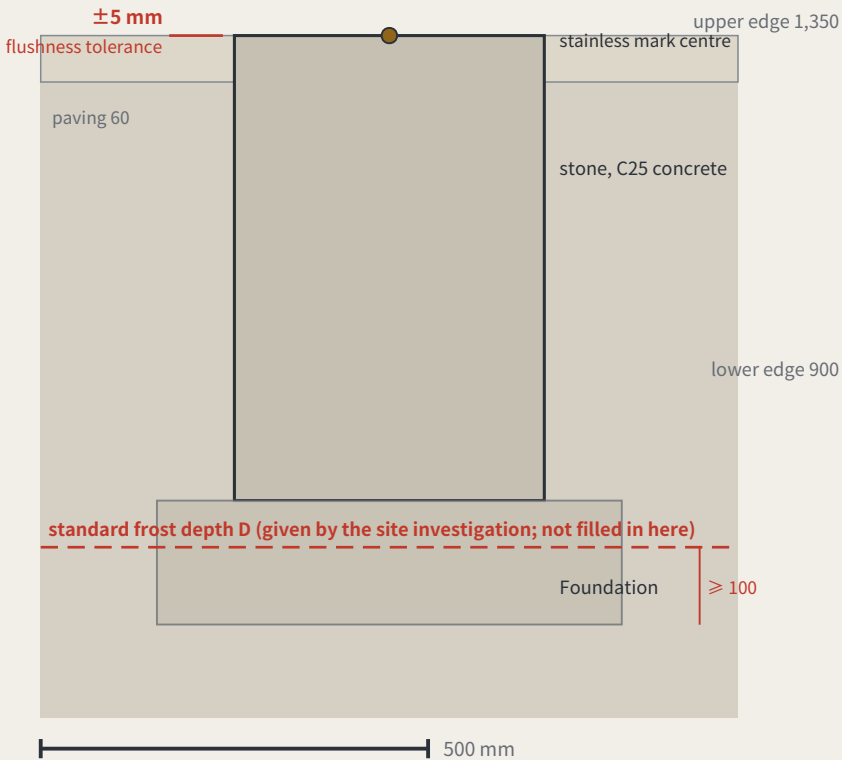


Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

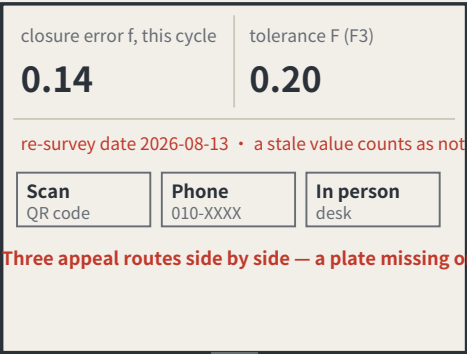
| Item                             | Design intent                                                           | Why this number                                                                                                                 | As measured |
|----------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------|
| The stone                        | 400 × 400 × 600 mm, C25 concrete, stainless mark centre in the top face | the mark centre is the point a benchmark_id refers to                                                                           |             |
| Stone top against the paving     | ≤ ±5 mm                                                                 | KIT-01 says flush with the ground and no trip hazard; without a tolerance, “flush” is an opinion                                |             |
| Depth to underside of foundation | ≥ the local standard frost depth D + 100 mm                             | D comes from the site investigation and is not filled in here — this package holds no verifiable frost-depth source for Haidian |             |
| Plate face size                  | 600 × 450 mm, raked 15°                                                 | the rake cuts glare and standing water; the face is replaceable without the post                                                |             |
| Plate lower / upper edge         | 900 mm / 1,350 mm                                                       | the band a seated and a standing reader share; P5’ s continuous step-free route should not end at a plate they cannot read      |             |
| Appeal panel                     | QR code, phone number and in-person address, all three side by side     | KIT-05: a QR code alone excludes anyone without a smartphone from appealing, which makes persona P4 unworkable                  |             |
| Tactile guidance strip           | 300 mm wide, stopping 300 mm short of the stone                         | the stop line leaves room to stand and take a reading rather than leading onto the stone                                        |             |
| Wheelchair turning space         | Ø1,500 mm, headroom ≥ 2,100 mm                                          | the turning space coincides with the reading position, so taking a reading needs no reversing first                             |             |
| Seat                             | seat 450 mm high, with armrests, within 2,000 mm of the plate           | KIT-03: taking a reading should not require standing and waiting, and the seat must be in sight of the plate                    |             |

II. Section



III. Plate elevation (post shown broken)

Specimen values; F is a dimensionless rate, read from scenario\_cards.json



the face panel is replaceable without touching the post

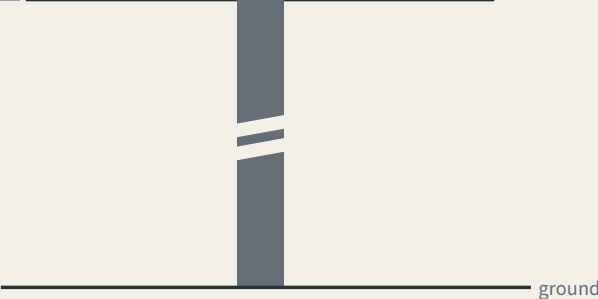
re-survey date 2026-08-13 • a stale value counts as not posted

Scan  
QR code

Phone  
010-XXXX

In person  
desk

Three appeal routes side by side — a plate missing one fails inspection



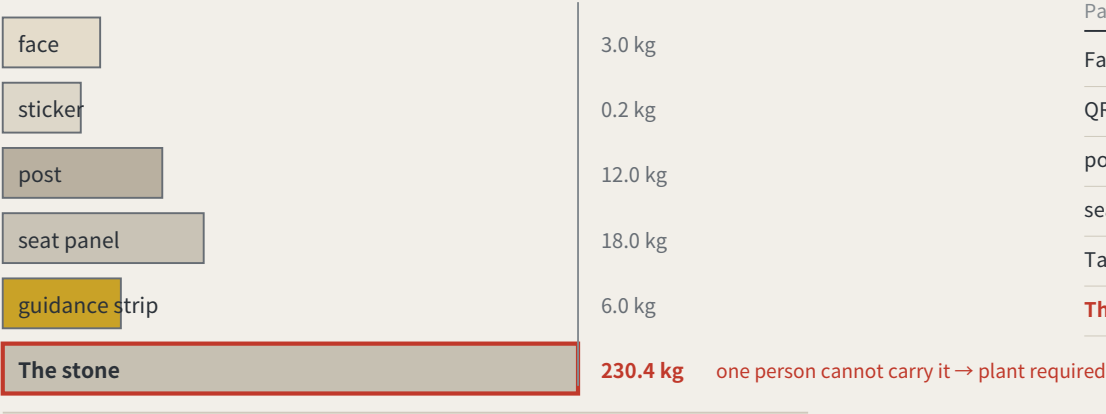
The whole proposal rests on one object, and not one dimension had appeared anywhere in the package — a component library without dimensions is a mood board. Here it is in plan, section and elevation, nine dimensions each with the reason it is that number; read the scale off the bars. The one figure left blank is the frost depth D: no verifiable source here, so the sheet gives the rule and not the number. The as-built column is empty too. Concept design intent; requires detailed design and engineering acceptance.

Maintenance: putting the wear on the parts that can be swapped

FIG.18

a type drawing; it carries no orientation

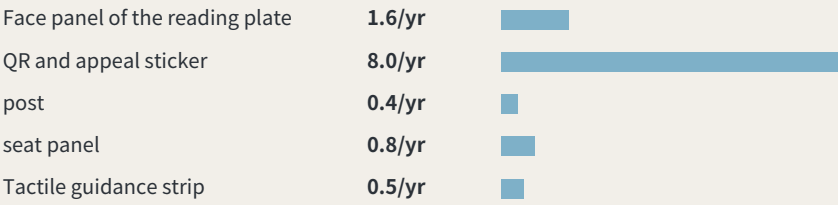
I. Order of removal: from what is swapped to what is never moved



One person is assumed to carry up to 23 kg. The stone, 400 × 400 × 600 mm of plain concrete at 2,400 kg/m<sup>3</sup>, works out at about 230 kg — the one part nobody can carry is exactly the one part that should never be replaced.

IV. What these intervals add up to

Summing the intervals above across 8 benchmarks: about 1.42 replacements per point per year, roughly 11 across the line, about 6 staff-hours a year. Those hours were not in the annual operating table — it counted re-survey sessions only. They are now: paid hours go from 133–214 h to 139–220 h, adding about CNY 680–1,473 a year. Maintenance is not a hidden cost, but it genuinely was not in the table.



Intervals and hours per swap are assumptions, not measurements; the stone is never replaced and is excluded.

II. What fails, how often it is replaced, and whether it touches the datum

| Part                            | How it fails                     | Replaced                                  | Mass    | Tool           | Touches datum |
|---------------------------------|----------------------------------|-------------------------------------------|---------|----------------|---------------|
| Face panel of the reading plate | stale values, scratching, fading | checked each cycle, replaced when damaged | 3.0 kg  | hex key        | no            |
| QR and appeal sticker           | the link dies, the surface wears | annual                                    | 0.2 kg  | none           | no            |
| post                            | impact, corrosion                | about 20 years, or after an impact        | 12.0 kg | socket wrench  | no            |
| seat panel                      | wear, graffiti                   | about 10 years                            | 18.0 kg | socket wrench  | no            |
| Tactile guidance strip          | abrasion, and repaving           | about 15 years, with the paving           | 6.0 kg  | paving tools   | no            |
| The stone                       | impact, excavation, settlement   | never replaced                            | 230 kg  | plant required | yes           |

III. After it fails: found → held → replaced → lost

|            |                                                                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Found      | any re-survey that finds the face stale or missing records it in that cycle<br>the plate’ s own rule — a stale value counts as not posted — is the detector                                                |
| Held       | the point is marked returned in datum red, shown at every third-order point in the area<br>bad news belongs where it happened (errata E50)                                                                 |
| Replaced   | a standard part is swapped without touching the stone or its mark centre<br>one person, one visit; heaviest part 18 kg                                                                                     |
| Stone lost | the height is re-established from the next order up; the closure record carries the break;<br>the id is retired and never reused<br>reusing an id would quietly join two different points into one history |

A monthly re-survey is a maintenance problem first, and this package had said nothing about what fails, how often, or who carries the replacement. One asymmetry organises the sheet: the stone is the only part that must never be replaced and the only one a person cannot carry (about 230 kg) — a benchmark that has moved is not the same benchmark. So the wear goes on the swappable parts. Intervals and masses are estimates, not measured.

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so  $f \leq F$  is always the passing test.

Act One: measure this open call first

|        |                                            |
|--------|--------------------------------------------|
| 228    | Merged proposals (git tree, authoritative) |
| 30.3%  | Machine-readable disclosure field empty    |
| 84 → 8 | Distinct strings written → model families  |

The “machine-readable” disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail,

and add one check to the gates.

Act Two: turn the same instrument on the site

|         |                                          |
|---------|------------------------------------------|
| 412.5 m | Provisional design area to surveyed park |
| 100%    | Agreement with the research area         |

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km<sup>2</sup> level agrees completely; the discrepancy is confined to the 11.4 km<sup>2</sup> level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

|       |                                        |
|-------|----------------------------------------|
| 8 / 8 | Cross-jurisdiction points on this belt |
|-------|----------------------------------------|

Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

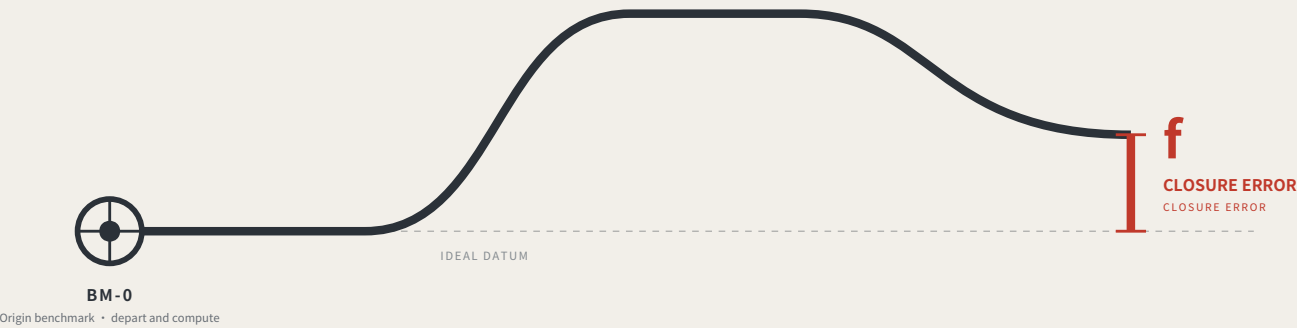


# THE LEVELING LINE

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## A city that publishes its own error.

A city that publishes its own error.



Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

jiangmuran • Open call for the Centennial Jing-Zhang AI Innovation Belt • Concept advice on provisional boundaries; not a substitute for statutory planning

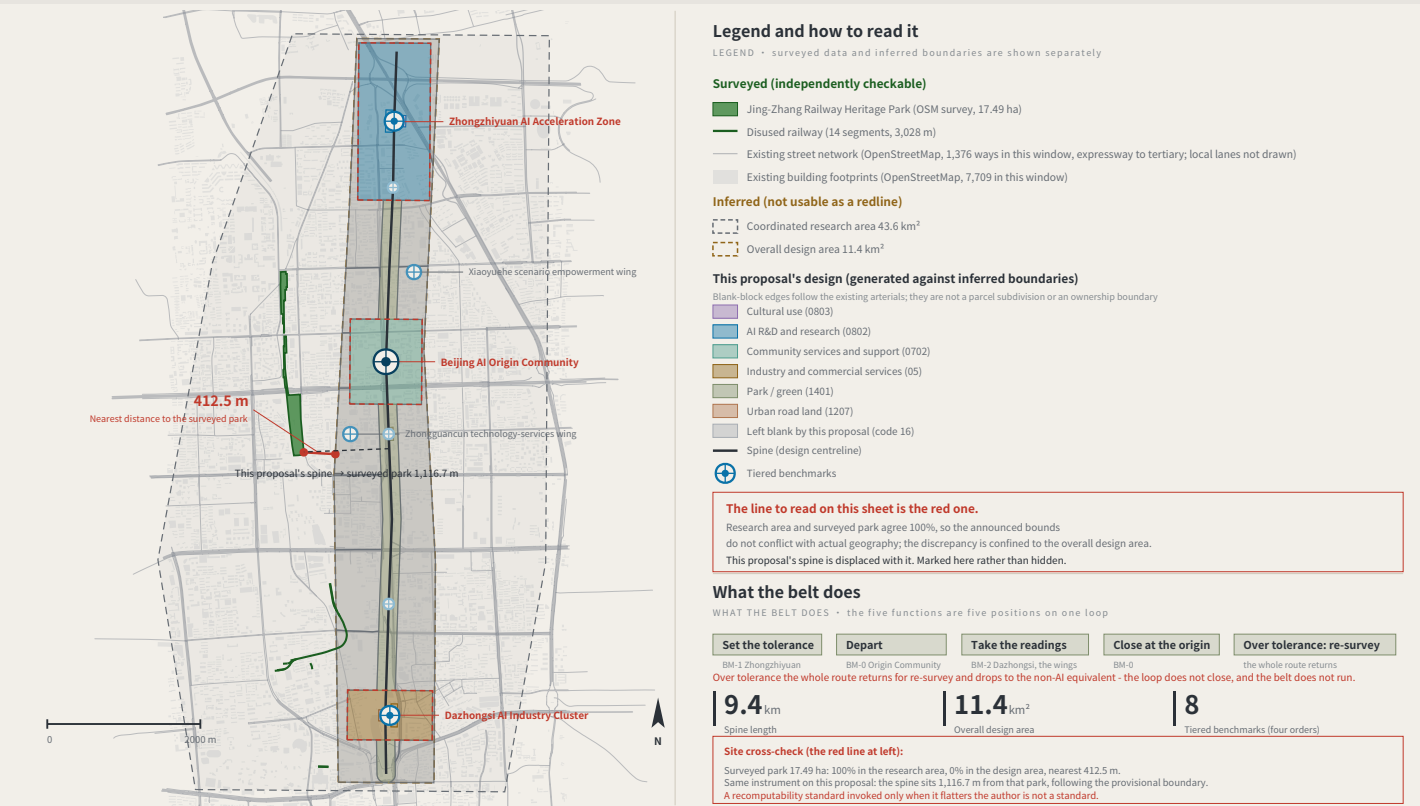
FIG.00

## OVERALL CONCEPT AND SITE OVERVIEW

FIG.01

Research area agrees 100% with the surveyed park; the gap is in the design area only, nearest 412.5 m — and this proposal's spine is displaced 1,116.7 m with it.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Surveyed data © OpenStreetMap contributors • ODbL 1.0 • Overpass API 2026-08-08. Script and conventions in visual/assets/osm\_reference.json; re-runnable. OSM is crowd-sourced and the park polygon may cover only the mapped section; the provisional boundary is itself inferred. This sheet reports the difference and does not adjudicate which is right.

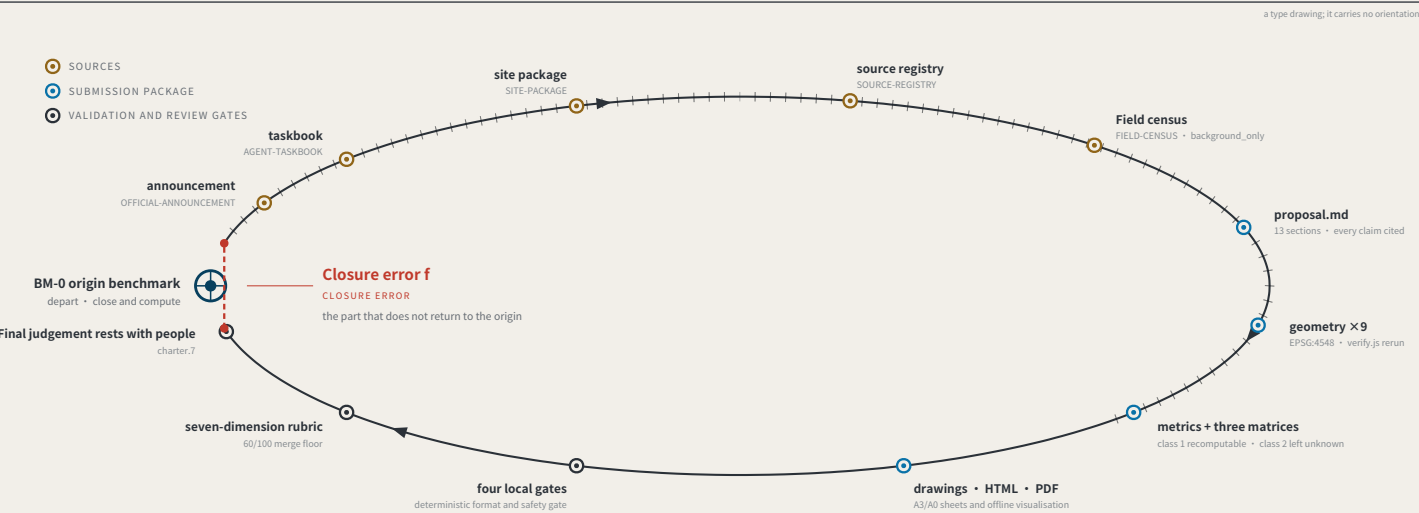
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Data to 2026-08-13 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

## EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

FIG.02



### Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-13 • 793 merged • 793/793 fetched • PR numbers past 1000

793

Merged proposals (git tree, authoritative)

The gallery index lists 507 at the same moment — 287 behind

18.2%

Machine-readable disclosure field empty

144 of 793 still hold the scaffold placeholder or are empty

228 → 9

Distinct strings written → actual model families

The GPT/Codex family alone is written 85 ways across 355 proposals

The "machine-readable" disclosure field is not machine-readable.

charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.

The fix is light: split "model" into an enumerated family plus free-text detail, and add one enumeration check.

This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field\_map.json and the first-round (184-file) snapshot agent\_declarations.json, all shipped with the package. The authoritative source is the GitHub git tree, not the gallery index, which lags. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; recompute before citing.

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Data to 2026-08-13 (sources.json; 1 not obtained)

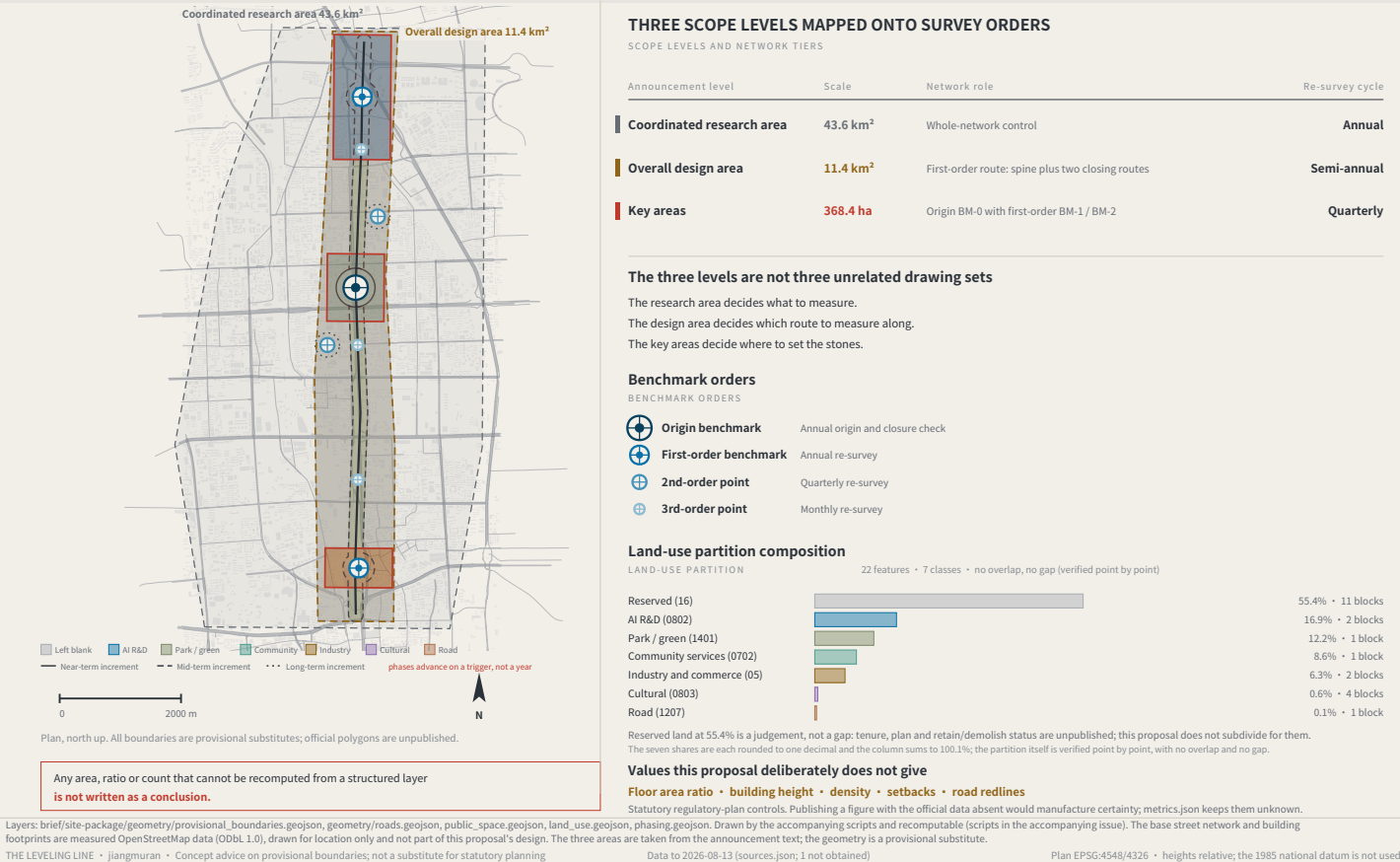
Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

## THREE SCOPE LEVELS AND NETWORK TIERS

FIG.03

7 land-use classes, no overlap and no gap, verified point by point; the 55.4% left blank is a judgement — tenure and control plans are unpublished.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

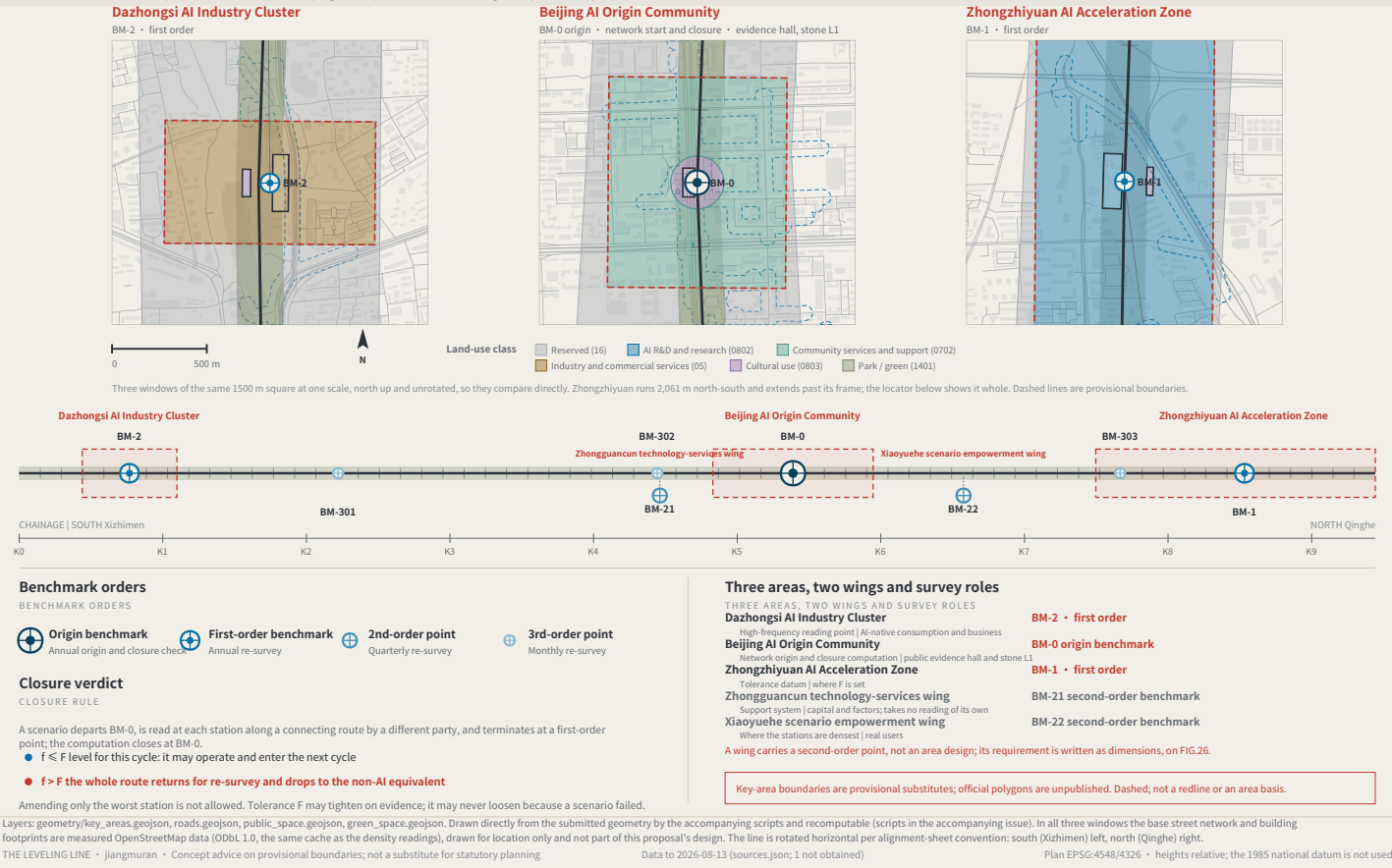


THREE AREAS, TWO WINGS AND BENCHMARK LAYOUT

FIG.04

Each area and wing carries a survey role — origin, tolerance, density; the blue dashes are the ten-minute walk catchment, by network distance, not radius.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

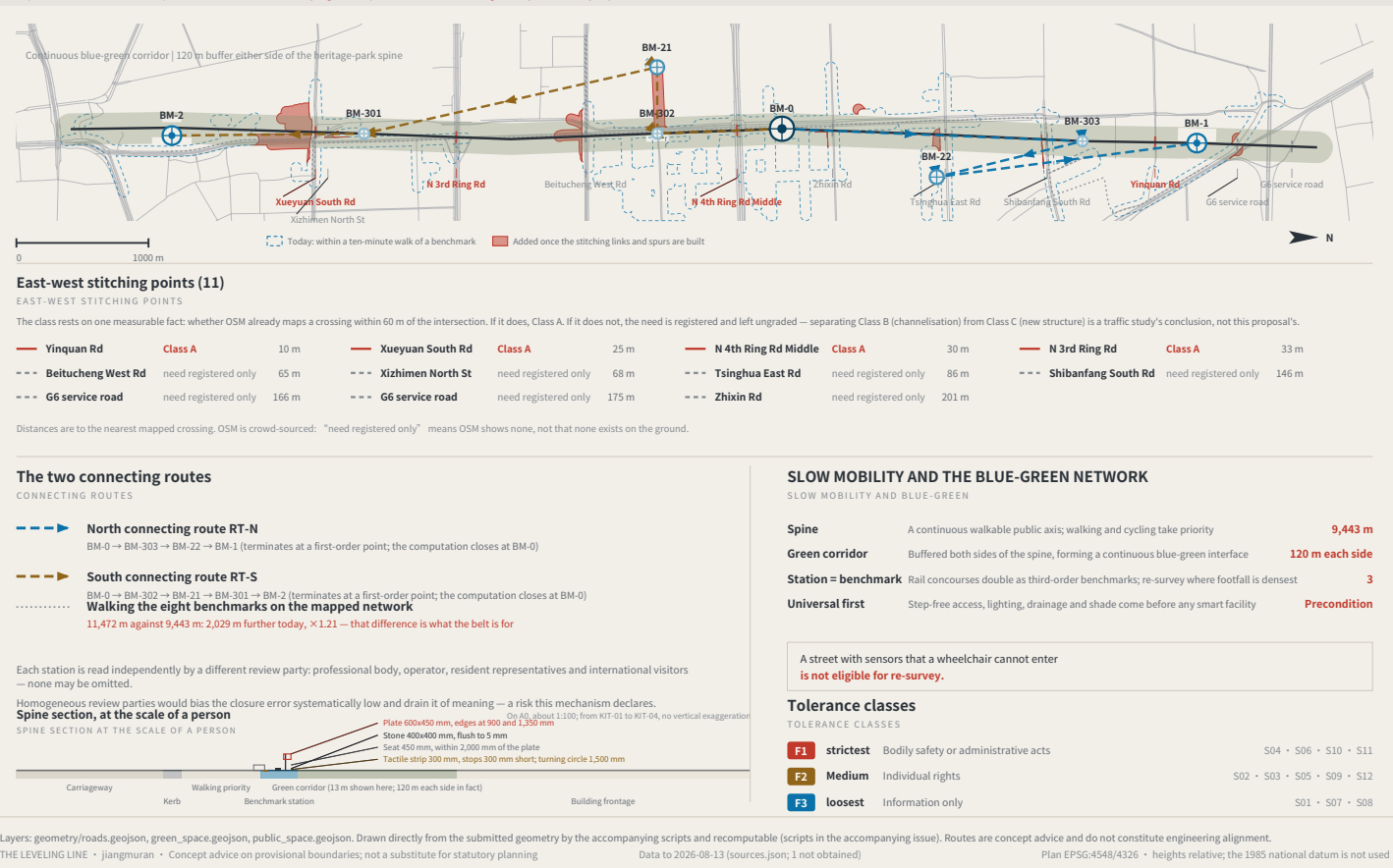


SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG.05

Walking the eight today is 11,472 m against 9,443 m drawn ( $\times 1.21$ ); with the stitching built, buildings within ten minutes of a benchmark go from 232 to 261.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

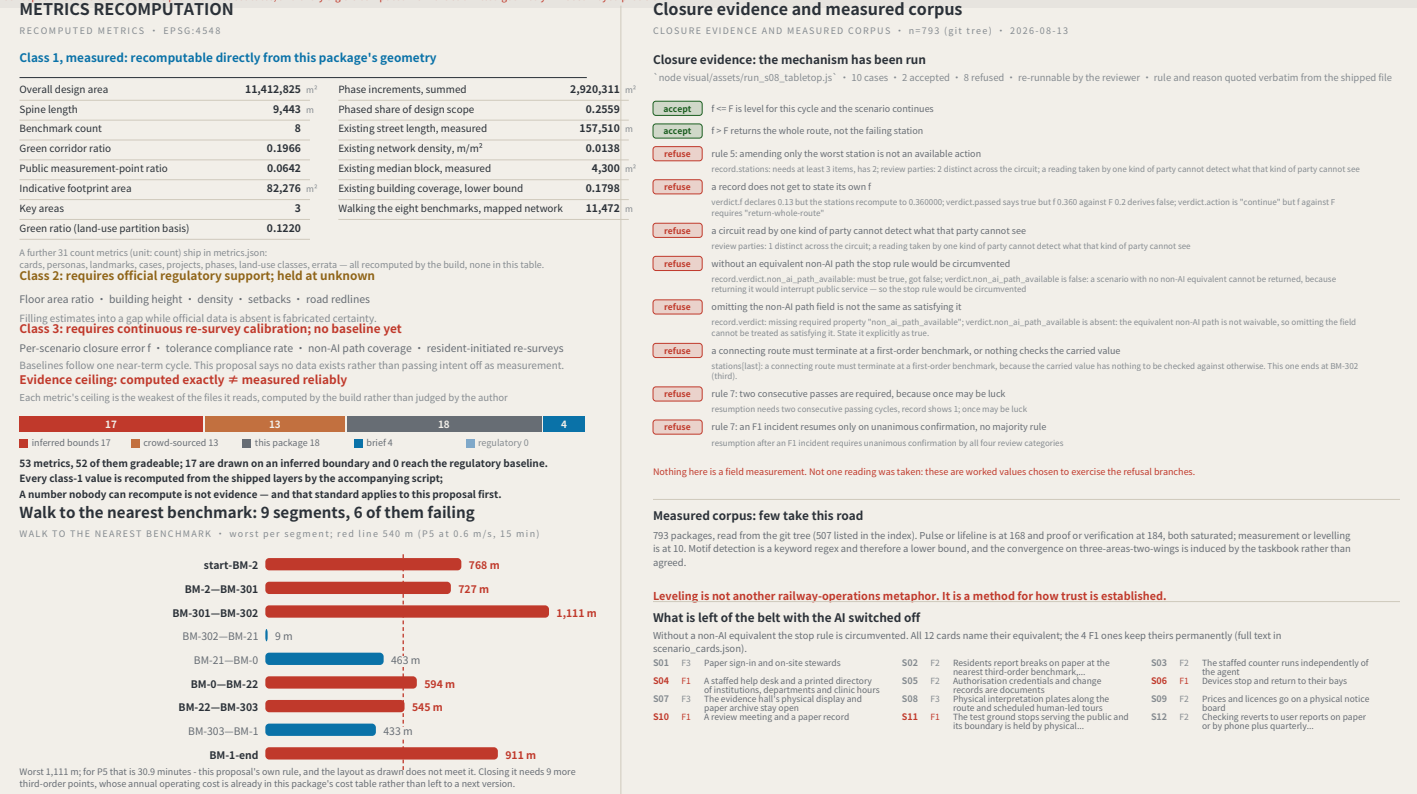


RECOMPUTED METRICS AND CLOSURE EVIDENCE

FIG.06

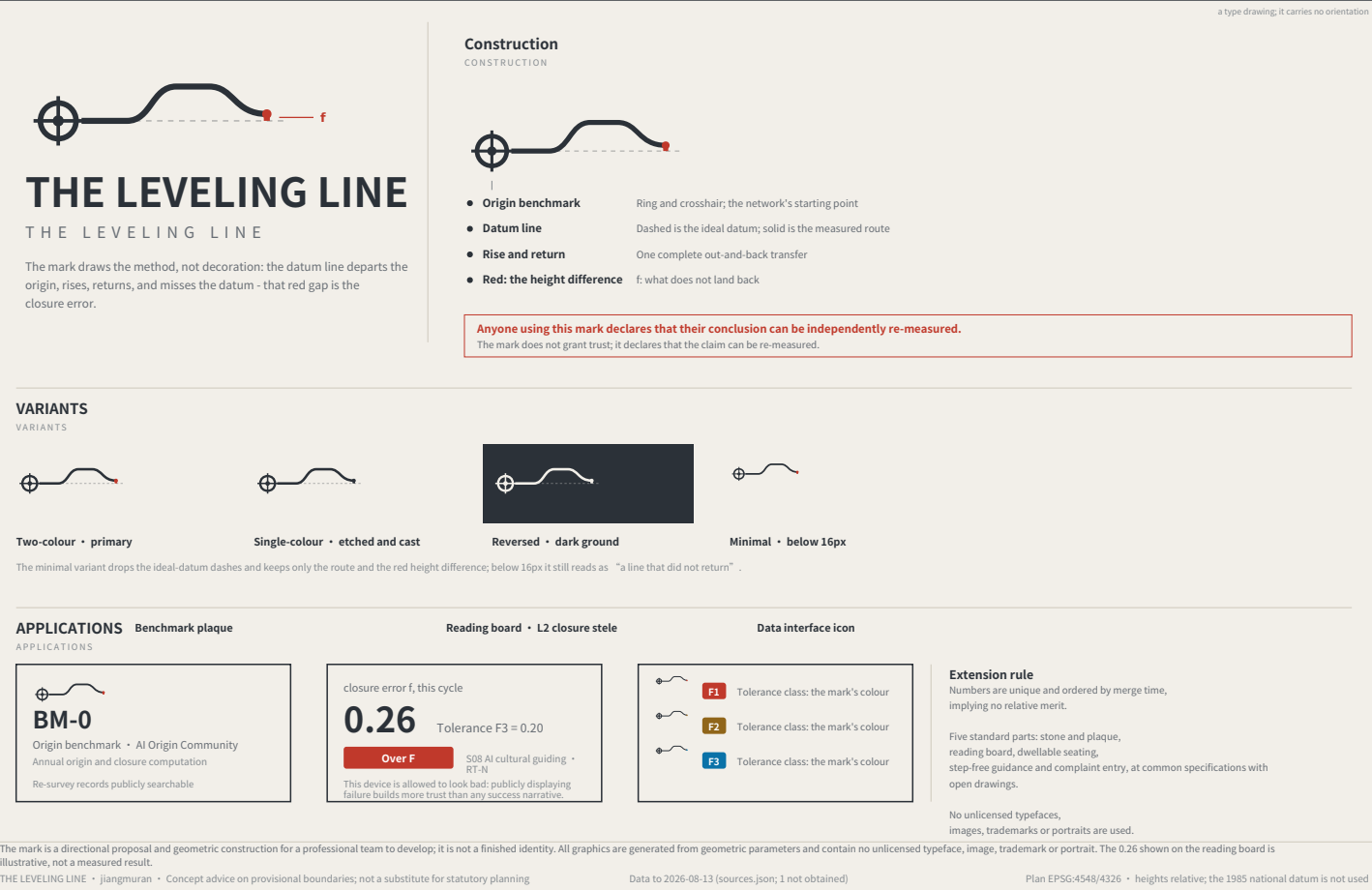
Run once: 10 cases, 2 accepted, 8 refused, reasons quoted from shipped files; 12 cards name the path that remains with the AI off.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

FIG.07



a type drawing; it carries no orientation

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

| Element        |   | Spatial carrier                       |   | Benchmark          |   | Re-survey item                              |
|----------------|---|---------------------------------------|---|--------------------|---|---------------------------------------------|
| Land and space | → | Key-area renewal parcels              | → | BM-1 / BM-0 / BM-2 | → | Delivery rate of promised space             |
| Compute        | → | Consolidated compute nodes            | → | BM-1               | → | Share of public compute open to application |
| Data           | → | Data-factor circulation venue         | → | BM-2               | → | Authorisation-chain completeness            |
| Scenarios      | → | Real-use venues in the two wings      | → | BM-21 / BM-22      | → | Enforceability of scenario exit             |
| Funding        | → | Zhongguancun technology-services wing | → | BM-21              | → | Deliberately blank – see the note below     |
| Talent         | → | Talent district and third places      | → | BM-0               | → | Re-survey of service satisfaction           |

The funding row has no re-survey item. That is a judgement, not an omission.

Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.

Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

| Id | Case                                   | Trust mechanism                                          | Re-measurable | Transferable point                                     |
|----|----------------------------------------|----------------------------------------------------------|---------------|--------------------------------------------------------|
| C1 | Algorithm registers                    | Public register of purpose, data, owner and appeal route | High          | The information base for a benchmark plaque            |
| C2 | Risk-tiered AI legislation             | Duties differentiated by risk class                      | Medium        | Supports the tiered setting of tolerance F             |
| C3 | Standardised testing frameworks        | Comparable reports from a common toolkit                 | High          | Gives re-survey its technical form                     |
| C4 | Algorithmic impact assessment practice | Mandatory ex-ante and ex-post review for high-risk uses  | red-high      | Supports the strictest tolerance for health scenarios  |
| C5 | Civic data-stewardship practice        | Data use decided by a citizen agenda                     | Medium        | Supports public re-survey rights at third-order points |
| C6 | Open-source reproducibility norms      | Conclusions must ship a re-runnable artifact             | High          | This package ships a runnable verifier                 |

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.

A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied; no marks used; no non-public data; and no fabricated company lists; investment figures or output values. Element-to-space correspondence is in geometry/land\_use, geojson and public\_space.geojson.

THE LEVELING LINE • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning

Data to 2026-08-13 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

a type drawing; it carries no orientation

Three AI pilgrimage landmarks

THREE LANDMARKS • spoken in engineering language; no spectacle

L1 The origin benchmark stone

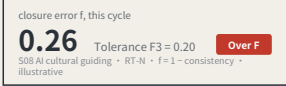
BM-0 • AI Origin Community



Every merged proposal's number is set into the paving – contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 • Zhongzhiyuan



A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. The value of this landmark is that it is allowed to look bad.

L3 return-to-datum point

BM-2 • Dazhongsi



An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

Kit of parts for public space: five standard components

KIT OF PARTS • common specifications and open drawings, so any new node joins in one language

BM-301

Third-order benchmark plaque

Monthly re-survey • unique number

Closure error f

0.14 Level, within tolerance

Next re-survey: March

1 Stone and plaque

The number is position, order and cycle

2 Reading board

Visible on site; no internet needed

3 Dwellable seating

With armrests: older people rely on them to stand

4 Step-free guidance

Read by the users themselves

5 Complaint entry

A non-scan method is mandatory, or the right does not exist for P4

Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured.

|             |                       |                                         |
|-------------|-----------------------|-----------------------------------------|
| BM-0        | Origin benchmark      | Annual origin and closure computation   |
| BM-1 / BM-2 | First-order benchmark | Annual re-survey                        |
| BM-2x       | 2nd-order point       | Quarterly re-survey                     |
| BM-3xx      | 3rd-order point       | Monthly re-survey                       |
| RT-N / RT-S | Closing route         | Semi-annual re-survey of the whole line |

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

|             |                   |                                  |                                 |
|-------------|-------------------|----------------------------------|---------------------------------|
| Monthly     | Community day     | third order BM-301/BM-302/BM-303 | P4 • P5 • P7                    |
| Quarterly   | Scenario open day | second order BM-21/BM-22         | P2 • P3                         |
| Semi-annual | Route re-survey   | first order, routes RT-N / RT-S  | P1 • P2                         |
| Annual      | Zeroing ceremony  | L3 return-to-datum point         | all four review parties present |

Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate  
International communication uses published readings as its material, never commitments.

The f=0.26 / f=0.14 on the reading boards are illustrative, not measured results. f is always a deviation: larger is worse, and f ≤ F is the passing test. Landmark siting requires heritage and engineering review; this proposal does not pre-empt it. The components are a directional proposal for a professional team to develop and are not construction drawings.

THE LEVELING LINE • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning

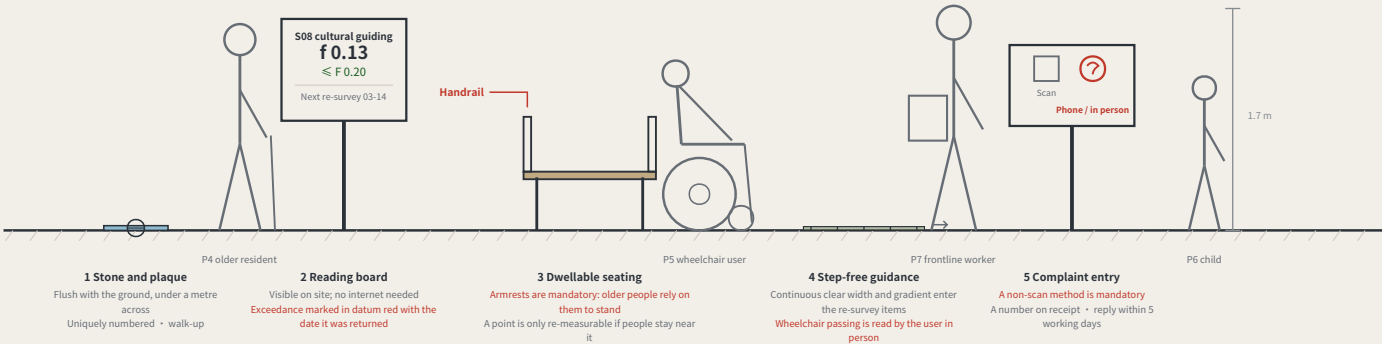
Data to 2026-08-13 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

a type drawing; it carries no orientation

What a benchmark looks like at eye level

A BENCHMARK AT EYE LEVEL • third-order BM-301 • elevation 1 m = 138 units



Two of the five standard components are hard constraints: seating must have armrests, and the complaint entry must offer a non-scan method. A rule you cannot see in the drawing is a rule the drawing does not contain.

The same kerb, after snow clearance

In winter the order becomes one real trade-off

|                                                                                     |                                                                                                         |                                                                                               |                                     |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Piled on the device band</b><br>Devices stop; S06 falls back to staffed transfer | <b>Piled on the pedestrian band</b><br>Pedestrians pushed into the carriageway - the worst of the three | <b>Piled on the kerbside parking band</b><br>The one item in the order that may be lost first | <b>This proposal takes this one</b> |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------|

Kerb allocation in priority order

KERB ALLOCATION IN PRIORITY ORDER • sequence and precedence only, no widths

|                                               |                                               |                                                                             |                                      |                                                                                    |                        |
|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------|------------------------|
| 1 Emergency and fire access<br>Never occupied | 2 Step-free boarding, turning<br>Not occupied | 3 Walking and dwelling<br>Never squeezed below the level-of-service reserve | 4 Public transport and cycle parking | 5 Device charging and standby<br>Sited only from what remains after the four above | 6 Kerbside car parking |
|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------|------------------------|

satisfied first → → allocated last

The order is itself a position: device charging sits behind pedestrians and accessibility, which means introducing devices must not be paid for with degraded walking conditions. Winter snow storage must be reserved at this stage alongside the device and pedestrian lanes — piled on the device lane the devices stop; piled on the pedestrian lane people are pushed toward the carriageway, which is worse.

The elevation is drawn at 1 m = 138 units. The figures are a scale reference and a statement about who is present, not a rendering. The section expresses sequence and precedence only and carries no widths — capacity must be computed on site from measured clear width, and a figure not derived from measurement is fabricated certainty. Component specifications are in geometry/public\_space.geojson and the scenario cards; positions await official boundaries and title verification.

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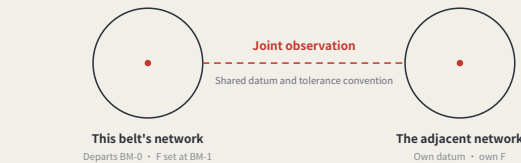
Data to 2026-08-13 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

a type drawing; it carries no orientation

An executable form of coordination: how two levelling networks join

AN EXECUTABLE FORM OF COORDINATION: HOW TWO LEVELLING NETWORKS JOIN



A joint observation produces a number, not an intention:

Each network reads the same public question set; the difference is the cross-network closure error. Within a jointly agreed tolerance, the two networks' records may be mutually recognised; over it, each re-surveys, and no declaration of partnership substitutes for a reading.

Nobody gives up authority: a joint observation asks only that conventions be comparable, not that organisations merge.

What each of the five partners exchanges, and what it does not

| Partner               | Stage                                            | Exchanges                                                                                       | Explicitly does not exchange                                                                                               | Precondition for the interface                                      |
|-----------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| BDA (Yizhuang)        | Volume production and high-level AI road testing | Mutual recognition of closure records across complementary speed domains                        | Not one admission: an F1 pass here admits a device to pedestrian density, not to a carriageway, and the reverse also holds | Both parties confirm the speed-domain split and test protocols      |
| Future Science City   | Corporate research institutes and engineering    | Scenario demand and real user density                                                           | No claim on IP; no scheduling on anyone's behalf                                                                           | Wing benchmarks built and a first baseline taken                    |
| Huairou Science City  | Large facilities and basic research              | The measurement method itself: how f is defined, cycle length, the statistics behind revising F | No scenario data is exchanged; the coordination is methodological, not applied                                             | The closure convention published and open to methodological review  |
| Beiwei Community      | The innovation community the taskbook names      | Reciprocal joint points; exchanged review parties                                               | This proposal does not know its positioning and does not set its benchmark order                                           | Each party names at least one publicly reachable joint point        |
| Beijing-Tianjin-Hebei | Cross-provincial coordination                    | Cross-provincial recognition of the tolerance convention                                        | No direct recognition of records; without a shared convention they are not comparable                                      | Tolerance definition and revision procedure agreed across provinces |

None of the five rows has been confirmed by the party named.

No internal plans are known, nobody is committed, no agreement is assumed; each row is an interface to evaluate independently.

Each partner's stage is described from publicly available general positioning and is unconfirmed by them. The speed-domain relationship in the first row must be settled against both parties' actual test protocols. This sheet proposes interfaces; it commits nobody and is no government determination.

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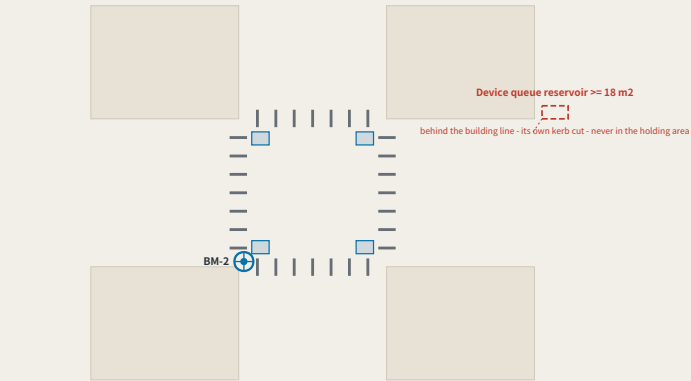


Four-quadrant pedestrian connection and device queue reservoir at Dazhongsi

FIG.12

I. Four-quadrant plan

FOUR-QUADRANT PLAN • TYPE APPLIED AT BM-2, NOT A SURVEY



Blue blocks are pedestrian holding areas, one per quadrant; the red dashed outline is the device queue reservoir, set behind the building line with its own kerb cut.  
If the four quadrants are not connected, devices and pedestrians must contend for the same holding area — which is why this district names it the most concrete spatial task it has.

2. Dimensions: design intent / as-measured

DIMENSIONS: DESIGN INTENT VS AS-MEASURED

| Item                                                                                         | Design intent            | As measured |
|----------------------------------------------------------------------------------------------|--------------------------|-------------|
| Pedestrian holding area, per quadrant<br>width $\geq$ 3.0 m x depth $\geq$ 4.0 m             | $\geq$ 12 m <sup>2</sup> |             |
| Crossing effective clear width<br>after deducting poles, covers and the furnishing strip     | $\geq$ 4.0 m             |             |
| Kerb ramp gradient<br>Barrier-Free Environment Construction Law and current codes            | $\leq$ 1:12              |             |
| device queue reservoir<br>set behind the building line, with its own kerb cut                | $\geq$ 18 m <sup>2</sup> |             |
| Reservoir to holding area, clear distance<br>must not cross any desire line                  | $\geq$ 2.0 m             |             |
| BM-2 stone to pedestrian edge<br>the stone is flush with the ground and is not a trip hazard | $\geq$ 0.6 m             |             |

**The right-hand column is deliberately empty.**  
Capacity is computed on site from measured clear width. A number filled in without measuring is manufactured certainty. Completed before any device is admitted.

3. Kerb allocation section (order and intended widths only)

KERB ALLOCATION SECTION

|                       |                              |                                                |                   |
|-----------------------|------------------------------|------------------------------------------------|-------------------|
| (1) Frontage<br>2.5 m | (2) Through walking<br>3.0 m | (3) Furnishing strip (BM-2 sits here)<br>1.8 m | (4) Kerb<br>0.7 m |
|-----------------------|------------------------------|------------------------------------------------|-------------------|

Building line to kerb: frontage, walking, furnishing, kerb. BM-2 sits in the furnishing strip, so a reading is never taken in the walking zone.

This is a type drawing applied at BM-2, not a survey of the existing junction. Every dimension is design intent and the as-measured column is deliberately empty: carrying capacity must be computed on site from measured effective clear width. Junction geometry, channelisation and signals rest with the road authority and a traffic study.  
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Data to 2026-08-13 (sources.json; 1 not obtained)  
Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

The three key areas in section, at one scale

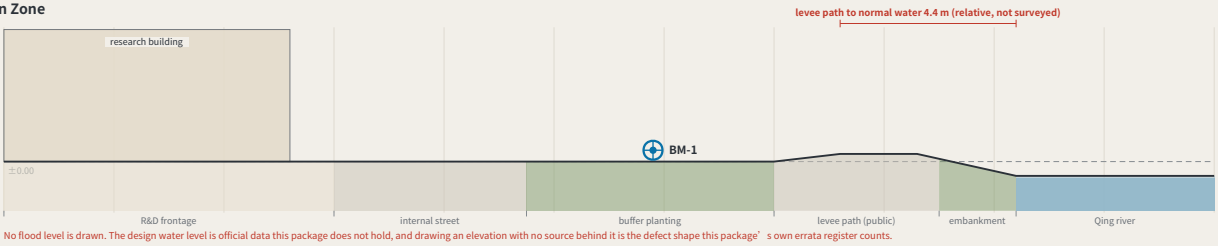
FIG.13

Zhongzhiyuan AI Acceleration Zone

$\pm 0.00$  = BM-1

192.9 ha • BM-1  
dominant use R&D 100%  
2 land-use classes inside

External arrival. The area is 100% single-use, so public access to the Qing river runs through the block or not at all — what this section sets out is a public route across the R&D frontage and onto the levee path.

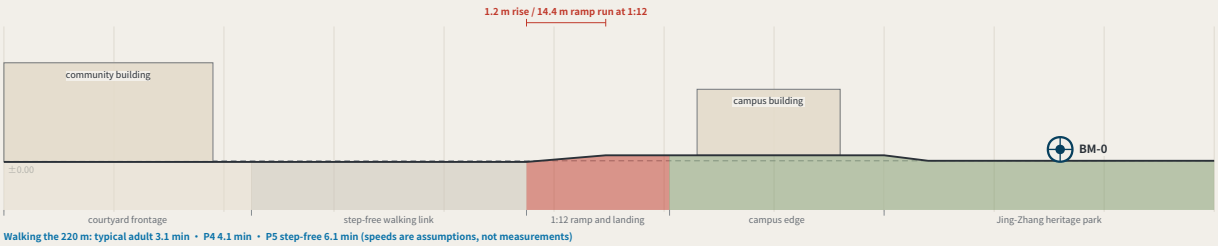


Beijing AI Origin Community

$\pm 0.00$  = BM-0

104.3 ha • BM-0  
dominant use community services 94%  
2 land-use classes inside

A 1.2 m grade change, and step-free continuity. The campus edge ends in steps, and P5 needs a continuous step-free route. The reason this section exists is to put a 1:12 ramp with a landing where the steps are.

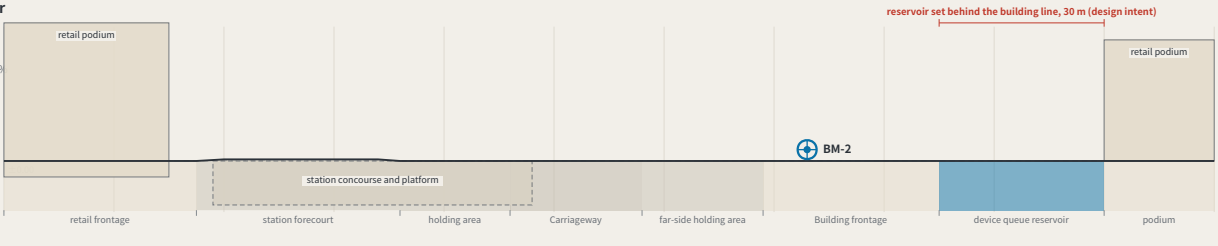


Dazhongsi AI Industry Cluster

$\pm 0.00$  = BM-2

72.0 ha • BM-2  
dominant use industry and commerce 99%  
2 land-use classes inside

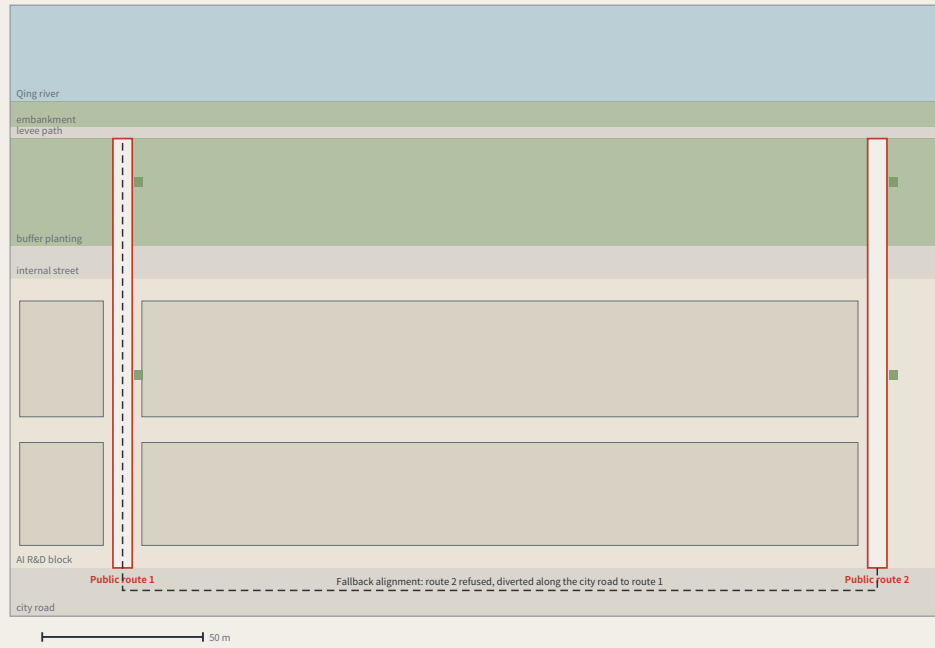
A station below grade and a queue behind the building line. The only section here with anything below ground, and the only one with a carriageway. The device queue reservoir sits behind the building line with its own kerb cut, clear of the crossing desire lines — the four-quadrant plan is FIG.12.



One horizontal scale, one datum convention, no vertical exaggeration. They are placed together so the charge that the three are one template used three times can be tested: one reaches a river across a 100% single-use block, one is step-free continuity over a 1.2 m rise, one is a station below grade with the device queue behind the building line.  $\pm 0.00$  is each area's own benchmark — absolute elevations need official levelling data this package does not hold, so they are not drawn. Concept design intent, not surveyed profiles.  
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Data to 2026-08-13 (sources.json; 1 not obtained)  
Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

The through-block public route to the Qing river

FIG.14



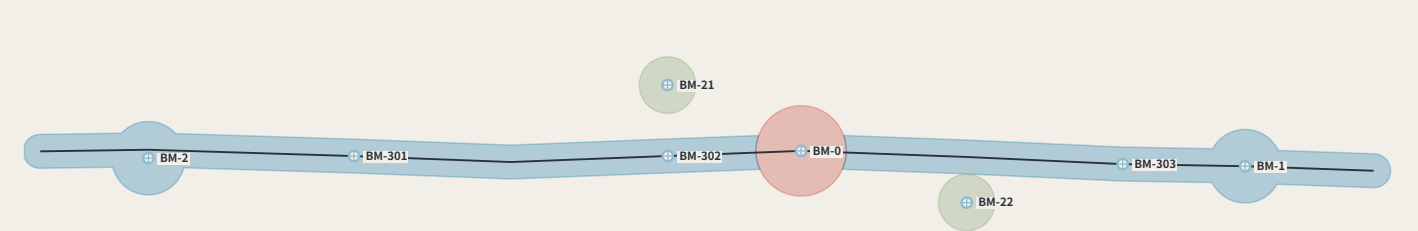
**The route along its length: city road to levee path**  
2.6x the scale of the plan • no vertical exaggeration • 133.5 m rising 1.4 m, an average of 1:95



FIG.13's section says this edge needs a public route across the R&D block onto the levee path, and no drawing showed it. This is that route: 6.0 m clear, centres at most 250 m apart (939.0 of them across a 4 m frontage), open 24 hours and ungated. The fallback alignment and the cost of a refusal are drawn. A type drawing, not a survey; dimensions are design intent, the as-measured column is empty; boundaries are provisional.  
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Data to 2026-08-13 (sources.json; 1 not obtained)  
Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Phasing: advanced by closure results, not by dates

FIG.15



Phases advance on closure results, not on years

| Phase | Increment | Spine added | Benchmarks this phase adds         | Condition that opens it (not a date)           | Re-surveys/yr | Annual running cost |
|-------|-----------|-------------|------------------------------------|------------------------------------------------|---------------|---------------------|
| Near  | 32.1 ha   | 0.64 km     | BM-0                               | none — no prior closure needed                 | 1             | CNY 2,400–7,740     |
| Mid   | 234.8 ha  | 8.80 km     | BM-1, BM-2, BM-301, BM-302, BM-303 | four near-term items done; $f \leq F \times 2$ | 39            | CNY 13,360–44,400   |
| Long  | 25.1 ha   | —           | BM-21, BM-22                       | mid phase: $f \leq F \times 2$                 | 47            | CNY 18,760–62,440   |

The table above phases the eight points that exist and their cumulative cost. FIG.21 shows that meeting the fifteen-minute rule needs 9 more third-order points (a further CNY 22,005–76,358 a year), falling by order into the mid and long phases — this is not the phasing of a network that already complies.

The first phase is a stone, a plate and one reading a year

The near-term increment is 32.1 ha and holds BM-0 alone: 1 re-survey a year, CNY 2,400–7,740 to run. That line is the answer to "this mechanism is too heavy" — it can start from one point, and that point costs what a sub-district office can approve by itself.

The condition is a number, not a year

The mid phase does not begin because a second year arrived. It begins because the four near-term items are complete and  $f \leq F$  for two consecutive cycles; the long phase the same. Once may be luck — the same rule this proposal writes at PATH-4 of the conversion path, applied to phasing. Fail it and the programme stays where it is, which makes the phasing table an exit table as well.

Which four are the four near-term items

The condition that opens the mid phase reads "the four near-term items complete" and nothing in the package says which four — and the proposal contains two different sets of four: the near-term projects R1–R4, and the ones needing no official data, R1–R3 and R8. The second set contains a long-term item, so reading the gate against it makes the gate circular. Taking the first: R1 datum stone L1 and the public evidence hall — one complete closure and its reading published within the first cycle R2 first public setting of the tolerance F — a first version of F1/F2/F3 each, posted R3 the four-week closure trial on cultural guidance S08 — closure recomputed and the consistency ratio published within four weeks R4 third-order benchmarks set out at communities and rail stations — one recorded reading at each of the three community points

Two naming errors this sheet found

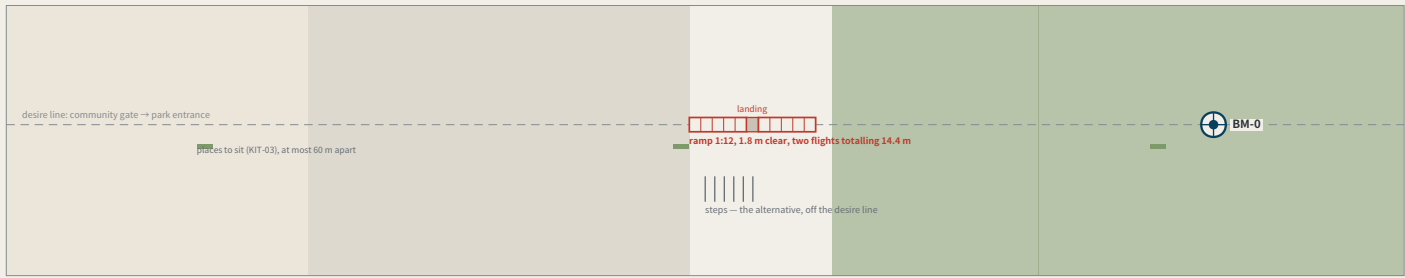
Dropping each benchmark into the phase polygons contradicts two of the phase names. The mid phase was named for "three first-order points" and contains two (BM-1, BM-2); the third is the datum BM-0, which is in the near phase. The long phase was named for "the full three-order network", and the three third-order community points already fall inside the mid phase — what the long phase actually adds is the two second-order wing access points. The names were written for the intent and never checked against the polygons. Both are corrected in the layer, and every cell of this table is computed from the geometry rather than typed.

phasing.geojson has shipped for a long time, been repaired once and enters the recompute path, and no drawing had ever shown it. This sheet draws the three increments and costs each phase's year with the same model as build\_operations, not a second set of coefficients. Phases open on closure results, not on years. The geometry is concept advice on provisional boundaries and must be recomputed when official ones are published.  
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Data to 2026-08-13 (sources.json; 1 not obtained)  
Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

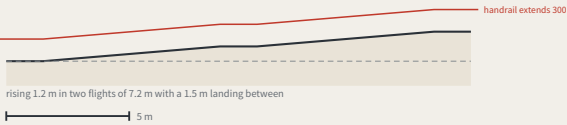
The step-free link from the origin community to the heritage park

FIG.17

I. Plan



II. The ramp along its length (no vertical exaggeration)



What offsetting the ramp would cost

The usual arrangement puts the steps on the desire line and offsets the ramp. Offset it by 22 m and whoever uses it walks 44 m further every time — out and back. For P0.6, at 5 m/s, that is 1.2 minutes a trip; at twice a week for a year, 4.6 km. Nobody on the steps walks a metre further for the ramp being on the line: steps are shorter, so they sit beside it.

Walking the 176 m

|               |         |  |
|---------------|---------|--|
| typical adult | 2.4 min |  |
| P4            | 3.3 min |  |
| P5            | 4.9 min |  |

speeds are assumptions, not measurements

Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

| Item                       | Design intent                                       | Why this number                                                                        | As measured |
|----------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------|-------------|
| Ramp gradient              | 1:12                                                | step-free throughout; the ramp is on the desire line and the steps are its alternative |             |
| Ramp clear width           | ≥ 1.8 m                                             | two people passing, or a wheelchair with someone alongside                             |             |
| landing                    | one every 750 mm of rise, at least 1.5 m long       | a 1.2 m rise split into two flights — there has to be somewhere to stop                |             |
| Handrail                   | both sides, extending at least 300 mm past each end | a handrail that stops at the top of the slope stops where it is still needed           |             |
| Level approach at each end | at least 1.5 m level                                | level ground before and after, so nobody has to turn on the slope                      |             |
| Spacing of places to sit   | ≤ 60 m                                              | standard component KIT-03, 450 mm high with armrests                                   |             |
| Steps                      | beside the ramp, off the desire line                | anyone who prefers steps is not sent around — steps are shorter than the ramp          |             |

FIG.13' s section says this place exists to put a 1:12 ramp where the steps are, and no drawing in the package had shown it. One design decision: the ramp sits on the desire line and the steps beside it — offset the ramp by 22 m and its user walks 44 m further every trip, while nobody on the steps walks a metre more. A type drawing, not a survey; dimensions are design intent and the as-built column is empty.

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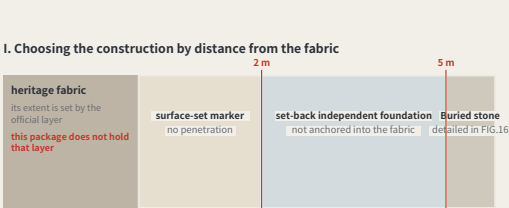
Data to 2026-08-13 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Monuments beside heritage fabric: the no-drilling construction and its setback rule

FIG.19

I. Choosing the construction by distance from the fabric



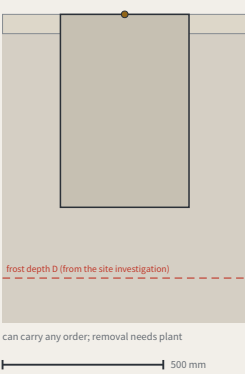
d = horizontal distance to the outer edge of the fabric

The rule is complete; its input is missing

Which line d is measured from depends on the officially published heritage protection area and its construction-control zone. This package does not hold that layer: each of the 3 shipped constraint features records it as a data gap in its own note, and this proposal does not infer it. So the sheet gives the rule, the thresholds and the three constructions, and leaves that line to whoever publishes it — guessing a protection boundary to make the drawing look complete would mean inventing the one thing here that is not this proposal' s to decide.

II. Two constructions, one scale

Buried: foundation below the frost line



Surface-set: bonded, no drilling, no excavation



III. What each construction costs

| Construction                        | Applies when | Foundation                                            | Orders it may carry            | Re-survey cycle                                         | What removal leaves                                  |
|-------------------------------------|--------------|-------------------------------------------------------|--------------------------------|---------------------------------------------------------|------------------------------------------------------|
| Buried stone                        | d ≥ 5 m      | below the standard frost depth D + 100 mm             | datum / first / second / third | per order                                               | plant to lift out; leaves an excavation              |
| set-back independent foundation     | 2 ≤ d < 5 m  | below frost line, set back ≥ 2 m, unanchored          | second / third                 | per order                                               | plant to lift out; leaves an excavation              |
| Surface-set marker (no penetration) | d < 2 m      | no foundation; bonded to existing paving, no drilling | third order only               | monthly, and tied to the nearest buried point each time | lifting it leaves only a bond mark, which cleans off |

The call is for a belt along a hundred-year-old railway; this proposal' s central act is sinking concrete. Three constructions, chosen by distance from the fabric, with the cost stated: without excavation there is no foundation below the frost line, so the surface-set marker may not carry first- or second-order duty. Where d is measured from depends on the official heritage layer, absent here and recorded as a gap in constraints.geojson.

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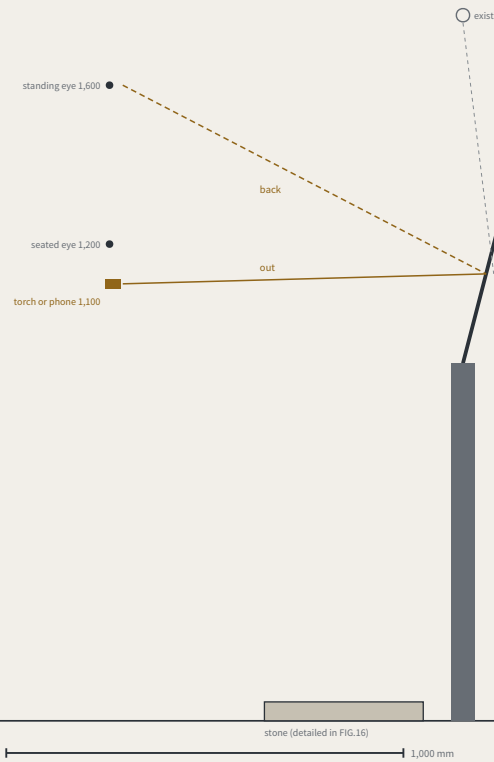
Data to 2026-08-13 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Reading after dark, without lighting the benchmark

FIG.20

I. Geometry: where the light comes from, and returns to



II. Three approaches, compared by how they fail

| Approach                                      | Needs power         | How it fails                                                                      | Light trespass               |
|-----------------------------------------------|---------------------|-----------------------------------------------------------------------------------|------------------------------|
| Retroreflective face, reader brings the light | no                  | the face gets dirty → visible in daylight → caught before it stops being readable | none                         |
| Low-level floodlight                          | yes, at every point | the lamp dies → the plate looks intact and cannot be read                         | yes, needs controlling       |
| Internally lit face                           | yes, at every point | the lamp dies → the whole face goes, and it still looks like a plate              | yes, and aimed at the reader |

The first is chosen not because it is cheap but because its failure is visible. When a lamp dies the plate still looks intact and can no longer be read — exactly the thing this proposal exists to prevent, a measurement whose failure cannot be seen. A dirty retroreflective face is simply dirty, visible in daylight to anyone walking past, well before it stops being readable.

plate face, raked 15°  
retroreflective film, returns light to its source  
a specular surface would bounce it straight into the reader — hence not specular

III. What is not lit

|                               |                                                                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No fixed luminaire            | the reader brings the light; no power supply, no luminaire to maintain                                                                                                  |
| The stone is not lit          | it sits flush with the ground; lighting it adds no legibility, only night-time glare                                                                                    |
| Nothing aimed upward          | no supplementary light may be emitted above the horizontal                                                                                                              |
| Nothing crossing the boundary | the added vertical illuminance at neighbouring dwelling windows must meet the local limit — that figure comes from the local standard, which this package does not hold |

Three third-order points are read monthly by residents, and much of the year those days fall after dark — and the package had never said how the plate is read at night. The obvious answer is a lamp, and it is the wrong one: it needs power, needs maintaining, throws light into the windows opposite, and when it dies the plate looks intact and cannot be read — a measurement whose failure is invisible, which is the thing this proposal exists to prevent. So the face is retroreflective with no fixed luminaire and the reader brings the light; a dirty face is simply dirty, in daylight.

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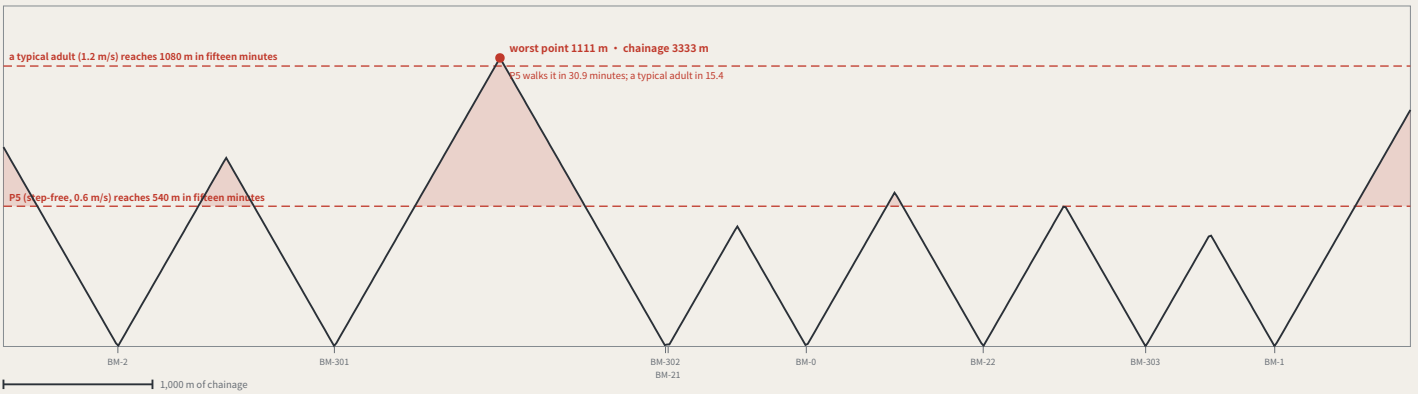
Data to 2026-08-13 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

How far the nearest benchmark actually is: this proposal's own rule, applied to itself

FIG.21

I. Distance to the nearest benchmark, along the spine



II. Turning that sentence into a spacing rule that can be checked

|                           |                                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Where the rule comes from | this proposal' s own repeated claim: a right of review you must walk fifteen minutes to reach has not been given                   |
| Interior spacing          | chainage between adjacent points ≤ 2 × 540 m = 1080 m                                                                              |
| Both ends of the line     | end of line to its nearest point ≤ 540 m                                                                                           |
| Which speed the rule uses | P5' s 0.6 m/s, not a typical adult' s 1.2 m/s — setting the rule by the faster figure sets it at a standard most people cannot use |

III. How far short the current layout falls

Six of the nine segments between the eight existing points fail. Closing the gap needs \*\*9 more third-order points\*\* (or 3 if the rule is only held to an average walker). They are not free: third-order points are read monthly, and every one of them enters the annual operating table. \*\*\*This sheet does not place them\*\* — putting 9 coordinates on a provisional substitute spine is the manufactured certainty this register counts. The rule ships; the positions wait for the official alignment.

This proposal argues repeatedly that a right of review you must walk fifteen minutes to reach has not been given, and its entire participation argument rests on anyone being able to walk up to a benchmark and take a reading — and nobody had ever measured that walk. Measured: the worst place on the line is 1111 m from the nearest benchmark, 5 minutes for P30.9, twice the limit this proposal holds others to. The rule ships as a spacing requirement; the positions wait for the official alignment.

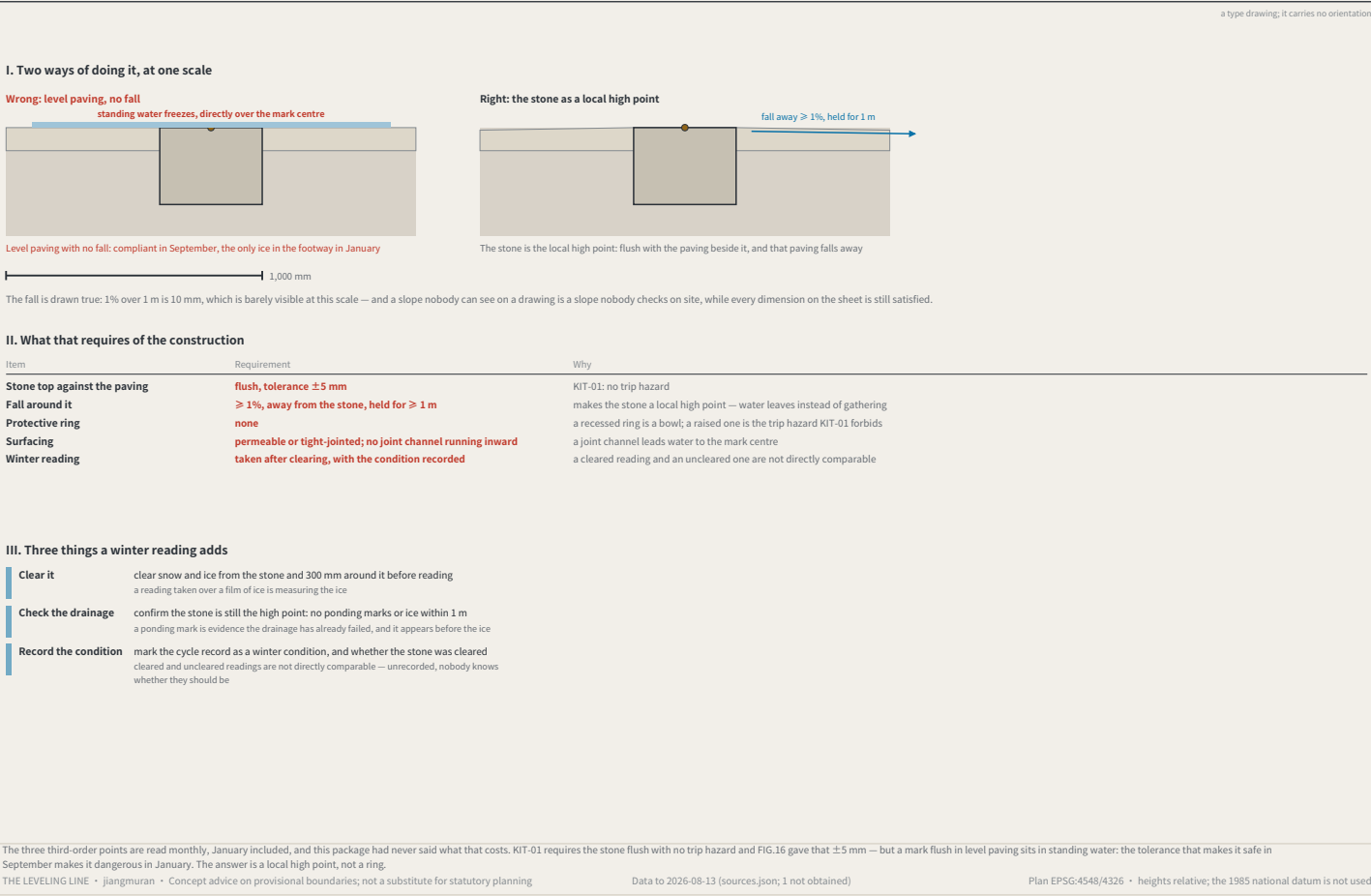
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Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

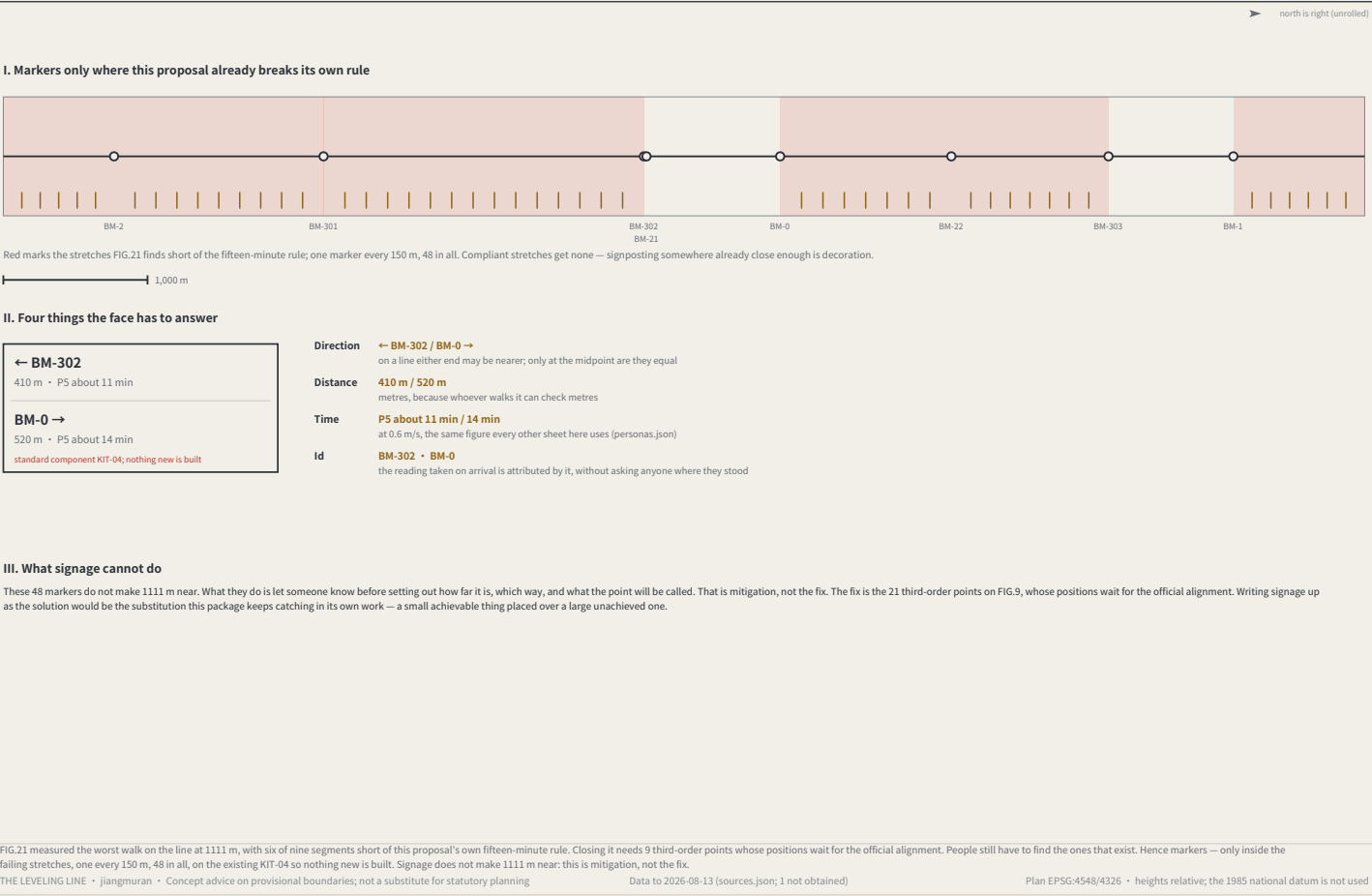
Winter: water, ice, and two of this package's rules in collision

FIG.22



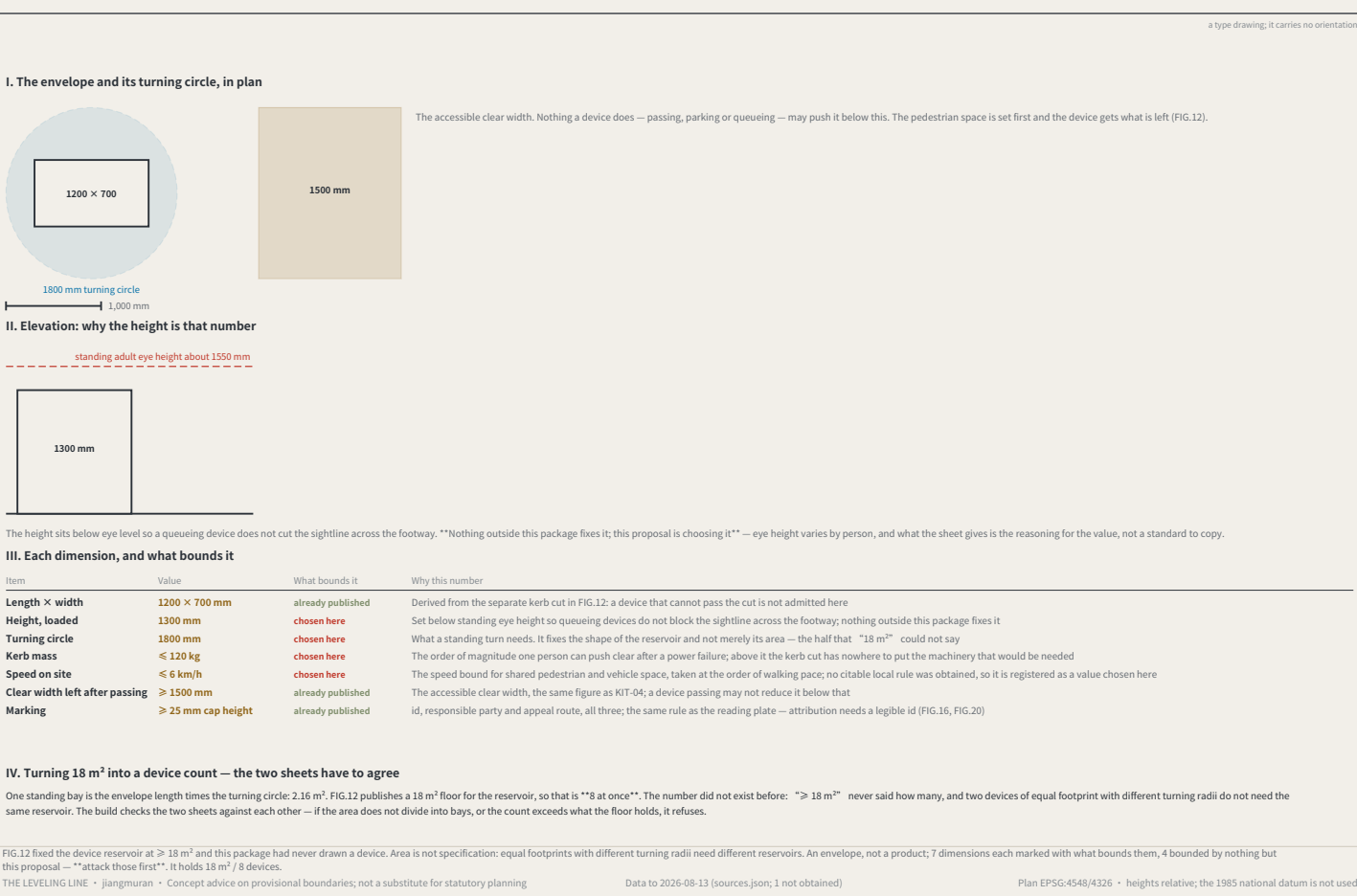
Finding the nearest benchmark: which way, and how far

FIG.23



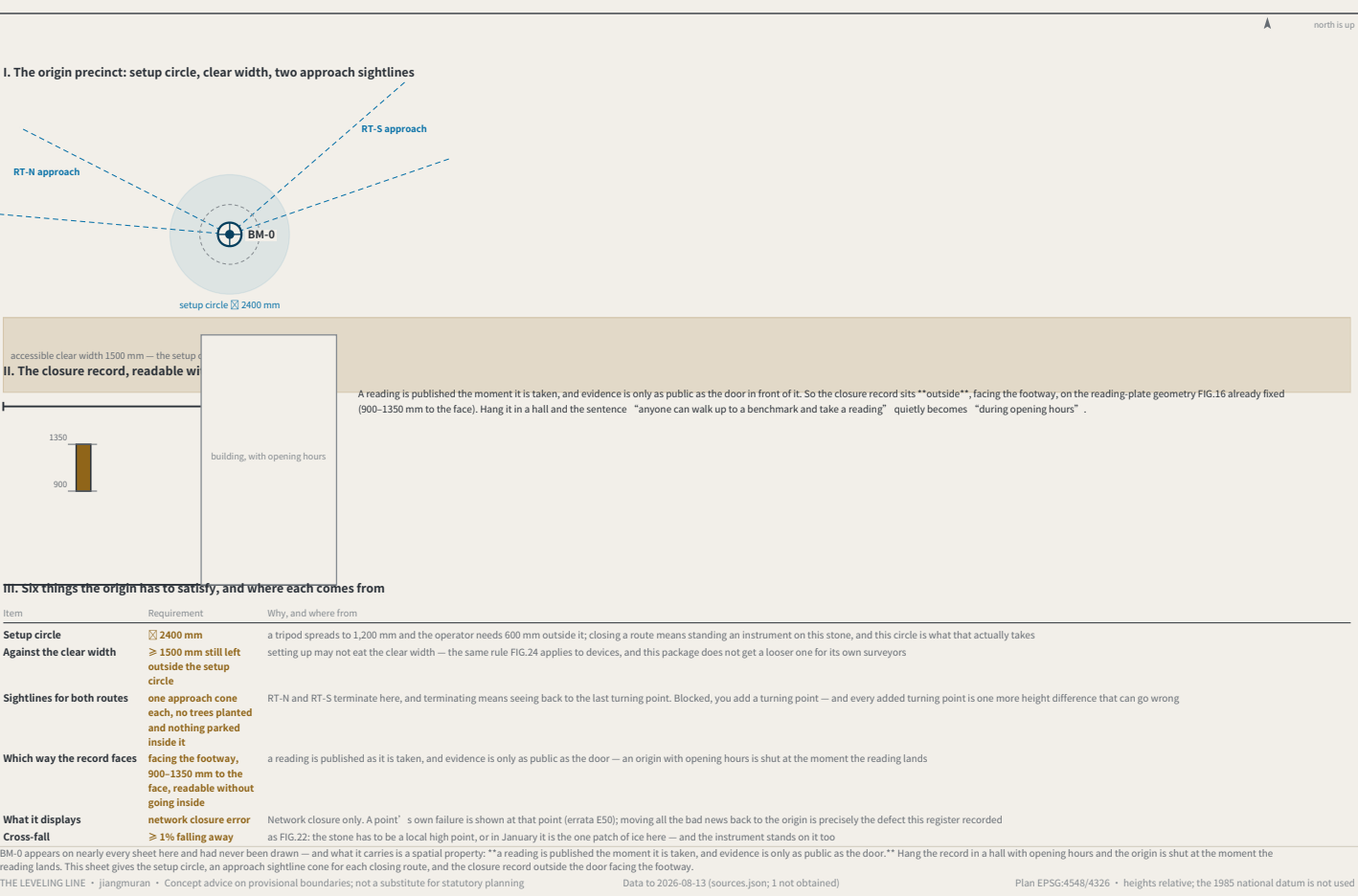
The device envelope: what exactly goes in those 18 m<sup>2</sup>

FIG.24



BM-0, the origin: the place both routes have to return to

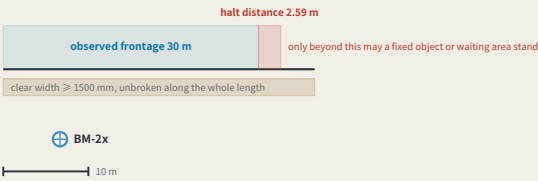
FIG.25



BM-2x, the second order: a scenario has to be stoppable, and that takes ground

FIG.26

I. The observed frontage and the halt distance, at one scale



II. This distance is computed, not decided

FIG.24 fixes site speed at ≤ 6 km/h = 1.667 m/s. Halt distance = reaction + braking = 1.667 × 1 + 1.667² ÷ (2 × 1.5) = 1.67 + 0.93 = \*\*2.59 m\*\*. \*\*The reaction time and the deceleration are chosen by this proposal and are not standards\*\* — registered as `A-DEVICE-002` on the register` s own terms, with derivation, consequence if wrong, closure condition and owner, in the same commit as this sheet. The formula ships with the drawing: substitute your own figures and this ground recomputes.

III. What makes second order differ from third

| Item               | Value                                                                   | Why                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The stone          | KIT-01, the same as a third-order point                                 | second and third order differ in cadence and route role, not in construction — a more important place does not get a special benchmark                                                        |
| Re-survey cycle    | quarterly                                                               | the wings are where measurement density comes from; readings are quarterly, priced in the same table as first-order annual and third-order monthly                                            |
| Observed frontage  | 30 m                                                                    | the length of kerb along which a scenario is actually observed. `Scenarios take their readings here` with no length is the same kind of sentence as `18 m² of reservoir` with no device in it |
| Halt distance      | 2.59 m                                                                  | computed from the site speed FIG.24 fixes, ≤ 6 km/h: 1 s of reaction covers 1.67 m, then decelerating at 1.5 m/s² covers 0.93 m                                                               |
| Clearance required | no fixed object and no waiting area within 2.59 m ahead of the frontage | a scenario that cannot stop is not a scenario but an incident; this ground is the physical form of the sentence `It can be stopped`                                                           |
| Clear width        | ≥ 1500 mm, unbroken                                                     | the same figure as FIG.24 and FIG.25 — observing, setting up an instrument and a device passing may none of them eat it                                                                       |

The two wings carry the second order — in the naming table, the cost table and the resource table — and had never been drawn. What differs from third order is not the stone (KIT-01, as everywhere) but that what runs here is \*\*a scenario against real users\*\*, so it has to be stoppable. From FIG.24` s ≤ 6 km/h the halt distance is 2.59 m, and no fixed object or waiting area may stand in that ground ahead of the frontage. \*\*The reaction time and deceleration are chosen by this proposal and registered as A-DEVICE-002.\*\*

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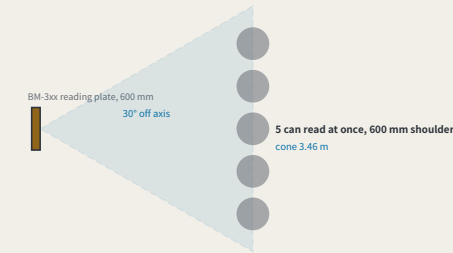
Data to 2026-08-13 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 · heights relative; the 1985 national datum is not used

Third order: the most numerous tier, and more than one reader

FIG.28

I. The legible cone, and how many can read at once



II. The part ground cannot solve

A class of 30 cannot read at once. 5 fit the cone, so 6 rounds, and at 40 s a reading that is about 4 minutes. \*\*The number is here because widening the ground does not solve it.\*\* No cone holds thirty people; the rest take turns, and turns cost time — a cost that, unwritten, gets discovered on site in the form of `this point is awkward`. The off-axis angle and the shoulder width are chosen by this proposal and registered as `A-READ-001`; the plate width, the viewing distance and the clear width come from FIG.16, FIG.27 and FIG.24 and are not fixed here.

clear width 1500 mm — a queue runs parallel to the kerb, never pushing into this

III. Standing rules for a third-order point

| Item                   | Value                                                                | Why                                                                                                                                                                           |
|------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Re-survey cycle        | monthly                                                              | third order is the most numerous tier and the one residents actually stand at                                                                                                 |
| Legible cone           | 3.46 m                                                               | a 600 mm plate (FIG.16) stays legible within 30° off axis, and at the distance FIG.27 fixes, 3.0 m, the cone opens to this width                                              |
| Readers at once        | 5                                                                    | cone ÷ a 600 mm shoulder. **This is this sheet answering the question FIG.27 left open** — that sheet fixed one person` s position                                            |
| A class of thirty      | 6 rounds, about 4 minutes                                            | past that count the extra people are not handled by ground but by taking turns — **and turns are a time cost, which belongs in writing rather than being discovered on site** |
| Which way a queue runs | parallel to the kerb, never perpendicular to the direction of travel | a perpendicular queue pushes its tail into the 1500 mm clear width; a parallel one does not                                                                                   |
| What is not provided   | no railing, no dedicated waiting area                                | a third-order point sits in a neighbourhood, and fencing it stops it being something you can read in passing; what this sheet gives is a standing rule, not a new structure   |

The origin and the second order each have a sheet; \*\*the third order is the tier nobody drew, and the one residents actually stand at\*\*. It also inherits what FIG.27 left open: that sheet` s 3.0 m distance is \*\*one person` s\*\* position. From a 600 mm plate and a 30° off-axis limit the cone is 3.46 m wide and 5 people read at once; a class of 30 takes 6 rounds, about 4 minutes — \*\*that part is solved by turns, not by ground\*\*.

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One reading: where a person stands, and whether they block the way

FIG.27

a type drawing; it carries no orientation

I. Where the person reading stands



clear width 1500 mm — the person reading may not stand inside it

II. This viewing distance comes from a ratio this package already published

This package has long used `legible height ≈ viewing distance ÷ 250` to check its own drawings, finding a minimum actual cap height of 4.73 mm, all passing. \*\*The same ratio had never been pointed at the plate.\*\* Pointed there, what it yields is not a type size but \*\*a place to stand\*\*: a 12 mm cap height gives 3.0 m, and 3.0 m is outside the 1500 mm clear route — the reader and the people walking past neither yield nor collide. At 6 mm the distance falls to 1.5 m and the reader stands in the middle of the path: \*\*the plate is still legible and the footway no longer works\*\*. The 12 mm cap height is chosen here and registered as `A-PLATE-001`. \*\*This package fixes another cap height too\*\*: FIG.24 requires a device marking at ≥ 25 mm, legible on the same ratio from 6.25 m. That is neither a contradiction nor an accident — a device marking identifies a machine \*\*coming toward you\*\*, and has to be read while there is still time to act; a reading plate is read standing in front of it. Different distances, different type. The relationship is stated because two cap heights with none stated between them is how a reader concludes one of them was picked carelessly. The ratio of 250 is not chosen here — it is the one this package already holds itself to.

III. The four steps of one reading, and the ground each takes

| Step       | What happens                                                                                                             | Why that dimension                                                                                                                                                             |
|------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Walk there | FIG.21 measured the worst walk on the line at 1,111 m, 5 minutes for P30.9 — this proposal does not satisfy its own rule | the cost of this step has been measured, and it came out failing                                                                                                               |
| Stand      | read from 3.0 m, standing clear of the through route                                                                     | 12 mm cap height × the 250 ratio = 3.0 m; at 6 mm it falls to 1.5 m and the reader stands in the middle of the path                                                            |
| Read       | read the figures on the plate and the benchmark_id                                                                       | the id lets the reading be attributed without asking anyone where they were standing                                                                                           |
| Disagree   | scan the appeal code, approaching to about 600 mm                                                                        | this step has to come close, so the code sits on the side away from the direction of travel — the one action that must intrude does so at the edge rather than across the flow |

The whole participation argument rests on one sentence — anyone can walk up to a benchmark and take a reading — and no sheet had drawn the four minutes that person spends standing there. Standing is not free: the ground they occupy is the ground everyone else walks through. Working back from the `legible height ≈ viewing distance ÷ 250` ratio this package already publishes, a 12 mm cap height gives 3.0 m, falling outside the 1500 mm clear route; at 6 mm the reader stands in the path. The cap height is chosen here and registered as A-PLATE-001.

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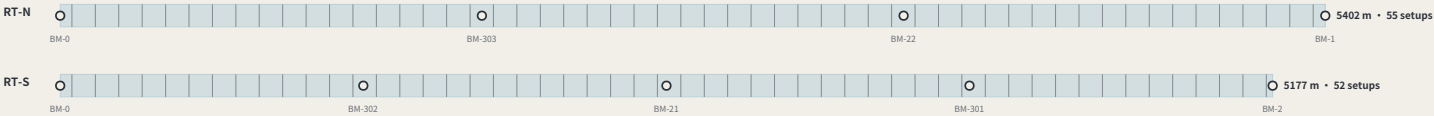
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The annual first-order closure: how long a route actually takes

FIG.29

north is right (unrolled)

I. The two closing routes, and how many setups each takes



Each vertical line is a setup, spaced by the 50 m sight — the instrument stands midway between two staves, so one setup advances the run by twice the sight. \*\*A closure is not one act but this many\*\*, and each is a place it can go wrong, which is exactly what the closure error measures.

1,000 m

II. The duration derived from geometry, against the band the cost table publishes

| Route | Length | Setups | Observing | Walk back | Total |
|-------|--------|--------|-----------|-----------|-------|
| RT-N  | 5402 m | 55     | 4.6 h     | 1.3 h     | 5.8 h |
| RT-S  | 5177 m | 52     | 4.3 h     | 1.2 h     | 5.5 h |

the band the cost table publishes for a first-order session: 4.0-6.0 h

III. These two numbers had never met

The 4.0-6.0 h for a first-order point in `operations.json` was written into the model by hand, on its author` s judgement, and nothing could check it. The route lengths are not hand-written: they come out of the shipped geometry. Divide by the sight into setups, multiply by the minutes a setup takes, add the walk back along the spine, and the result is 5.5-5.8 h — \*\*inside that band, and tight against its upper edge\*\*. The build asserts it: lengthen the geometry or edit the band and the two stop being compatible and the build stops. The sight distance and the minutes per setup are chosen by this proposal and registered as `A-SURVEY-001`; the route lengths, the walking speed and the hour band are all already published here.

The first order was the last tier without a sheet, and it is not really a place but \*\*an event\*\*: once a year a closing route is run, and this proposal` s measure of trust comes out of that run. From the shipped route lengths, a 50 m sight and 5 minutes a setup, the two routes need 52-55 setups and 5.5-5.8 h — \*\*inside the 4.0-6.0 h the cost table sets by hand\*\* — and the build asserts the two stay compatible.

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The year: where forty-seven readings fall

FIG.30



FIG.29 priced “one” closure and the cost table prices “a year” — and no sheet had drawn the year. Drawn, it shows that with every cadence starting in the same month the peak is 83.5 h against a 7.5 h trough, eleven to one: a team sized for the average cannot do the peak month, and one sized for the peak is idle the rest of the year. Separating the three annual readings drops it to 31.5 h with “no change to the work at all”. This is a calendar problem, not a resourcing one.

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The other year: the hours this network asks of people it does not pay

FIG.31



FIG.30 flattened “paid” hours and said the quarterly points need not move. Volunteers attend only the second and third order — precisely the tier left alone — so the volunteer year stayed at 31.0 h against 15.0 h. Separating them brings it to 23.0 h at a cost of 2.0 h on the paid peak. “Both cannot be minimised at once, and this proposal takes the volunteer peak”: a paid peak is a procurement problem, a volunteer peak is a participation failure.

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How many people: the cheapest roster is the one that empties the instrument

FIG.32

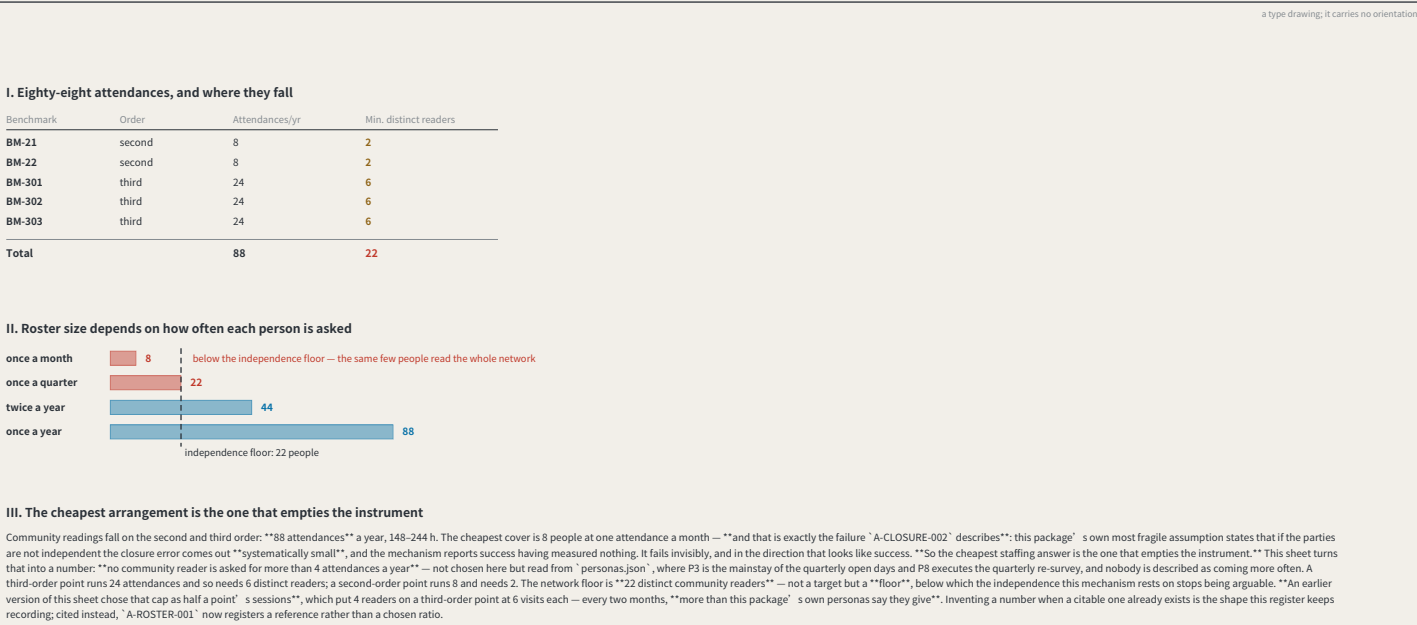


FIG.31 scheduled volunteer “hours” and never asked how many “people” that is. The year needs 88 community attendances, and the cheapest cover is 8 people once a month — “which is exactly the failure A-CLOSURE-002 sets out”: parties that are not independent make the closure error systematically small, and the mechanism reports success having measured nothing. The rule: nobody attends more than half a benchmark’s readings in a year, giving a network floor of 22 people.

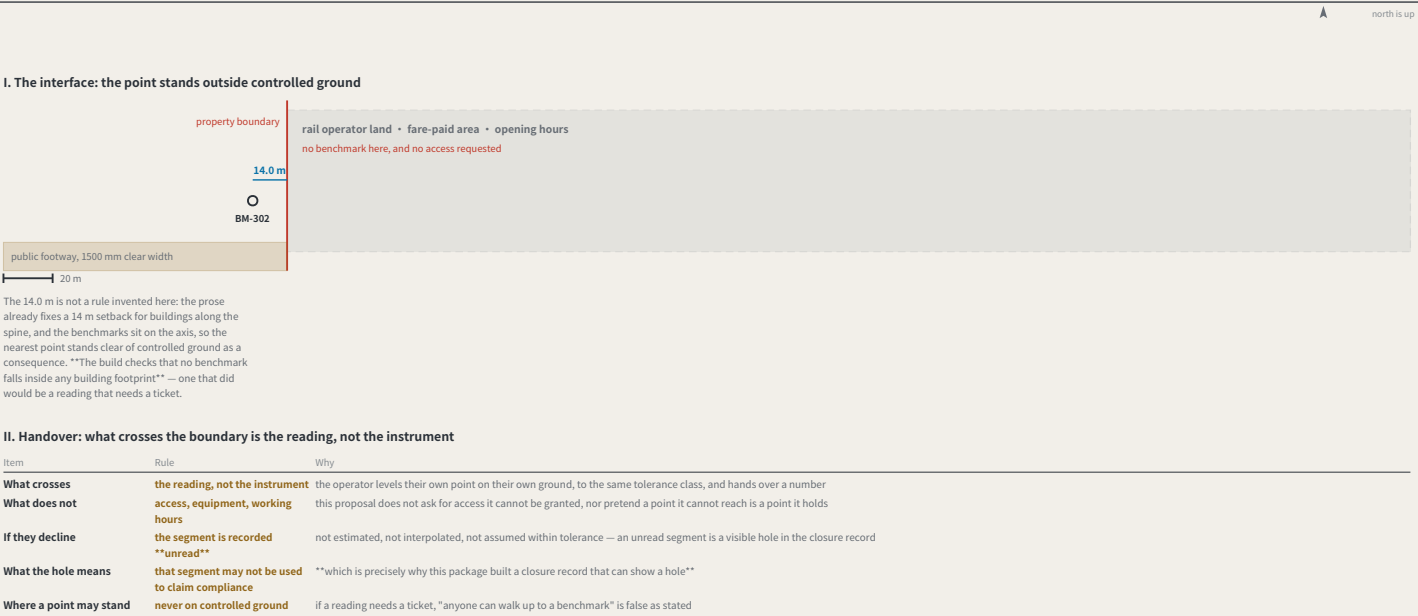
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Dazhongsi: where the belt meets ground this network does not own

FIG.33



Every other sheet assumes this network reaches its own points; at the station that ends — the landowner is not the city. Three rules: “no benchmark on controlled ground” (BM-302 stands 14.0 m clear of the hub, checked at build time); “what crosses is the reading, not the instrument”; and “if the operator declines the segment is recorded unread” — not estimated, not interpolated, a visible hole in the closure record.

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